A Systematic Analysis of Linguistic Features in AI-Generated Text Detection Across Domains and Models

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Abstract

Interpretable linguistic features offer a promising approach for explaining *why* a given text appears machine-generated, particularly for non-expert users. However, existing findings on which features reliably indicate LLM-generated text remain fragmented across feature sets, models, and text domains. To address this gap, we conduct a large-scale empirical study assessing the robustness of linguistic signals for characterizing AI-generated text. Our analysis covers 284 interpretable linguistic features across outputs from 27 LLMs and ten text domains under cross-model and cross-domain generalization settings. We show that classifiers based solely on linguistic features can reliably distinguish AI-generated from human-written text. However, many previously proposed indicators prove strongly context-dependent, with the exception of measures of lexical richness, which remain robust signals across model families and text domains. These results demonstrate which linguistic signals generalize across contexts and provide a foundation for more reliable, interpretable analyses of AI-generated language.

1 Introduction

Large Language Models (LLMs) are becoming increasingly adept at producing fluent and stylistically *human-like* text. The societal implications of this development are significant: the ease of producing large-scale synthetic text amplifies the risk of misinformation while simultaneously challenging cues used to assess authorship and factuality (Srivastava, 2025). This creates a growing need for *interpretable explanations* of why a text appears machine-generated, particularly in settings where unreliable or biased detectors can have significant consequences (Jiang et al., 2024).

Most existing approaches explain why a text appears machine-generated by interpreting the decisions of black-box classifiers (Zellers et al., 2019; Li et al., 2024a; Guo et al., 2025). However, such explanations are often difficult to interpret for non-experts (Ji, 2009) and typically provide only local, example-based insights into the properties of generated text (Koike et al., 2025). In response, a growing line of research focuses directly on *interpretable linguistic features*, identifying stylometric and linguistic differences between human and AI-generated writing, such as lexical diversity and syntactic regularity (Muñoz-Ortiz et al., 2024; Opara, 2024; Reinhart et al., 2025). While these studies provide promising evidence that linguistic features can characterize AI-generated text, the findings remain fragmented across feature sets, LLMs, and text domains. Consequently, it remains unclear which signals reflect general properties of generated language and which arise from particular datasets or model settings (Kehkashan et al., 2025).

In this work, we take a systematic approach to clarify the fragmented evidence on linguistic signals of AI-generated text. To this end, we conduct a large-scale empirical study of AI-authorship detection across 27 LLMs and 10 text domains from the MAGE benchmark (Li et al., 2024a). Our analysis uses a comprehensive set of interpretable linguistic features (284 in total), aggregating and extending feature groups previously studied in stylometry, readability analysis, and AI-text detection. With a focus on realistic cross-model and cross-domain generalization settings, where detectors must operate on unseen models or text types, we answer three central questions:

RQ1 How robust are linguistic features for distinguishing LLM-generated from human-authored text across models and domains?

RQ2 Which linguistic features consistently contribute to distinguishing AI-generated from human-written text across models and domains?

RQ3 How do linguistic signals of AI-generated text vary across model families and text domains?

First, we show that classifiers based solely on linguistic features achieve strong performance across models and domains (§4.1), demonstrating that interpretable linguistic signals can reliably distinguish human- and AI-generated text. Second, systematic feature-ablation experiments reveal that many previously reported linguistic indicators vary substantially across models and domains, with measures of lexical richness emerging as the most consistently informative group across contexts (§4.2). Finally, linguistic signals vary systematically across both model families and text domains, with different models exhibiting distinct stylometric patterns (§4.3). Taken together, this work provides a systematic evidence that linguistic characteristics of AI-generated text exhibit both robust and context-dependent patterns: while some signals, such as lexical richness, generalize across models and domains, others depend on specific generation settings. This highlights the importance of evaluating linguistic explanations under realistic cross-model and cross-domain conditions.

2 Related Work

Much of the literature on AI-generated text focuses on the task of **automatic detection** (Gehrmann et al., 2019; Li et al., 2025; Zhang et al., 2024; Guo et al., 2024; Ji et al., 2024, inter alia), often relying on black-box systems (Zellers et al., 2019; Guo et al., 2025). Efforts to improve interpretability typically rely on post-hoc explanation methods, such as feature attribution (Ji, 2009) or example-based explanations (Koike et al., 2025). In parallel, several large-scale benchmarks and shared tasks have been proposed to evaluate detection performance across datasets and models (Koike et al., 2024; Li et al., 2024b; He et al., 2024; Wang et al., 2024; Li et al., 2024a, inter alia). However, the primary focus of this strand of research remains improving detection accuracy rather than interpretability.

A smaller but growing line of research instead examines **interpretable linguistic signals** that distinguish AI-generated from human-written language. For example, Muñoz-Ortiz et al. (2024) identify morphosyntactic and semantic indicators that differentiate human- and AI-authored texts in the news domain, while Dönmez et al. (2025) analyze LLM-generated counterarguments. Reinhart et al. (2025) employ Biber's 67-feature tagset to explore linguistic patterns in generated text, Opara (2024) propose a stylometric detection approach based on handcrafted indicators, and Doughman et al. (2025) uses a small set of linguistic features (POS/NER, readability, and lexical features) for post-hoc analysis of LLM-based detection. These studies demonstrate that interpretable linguistic features can reveal systematic differences between human and AI-generated texts. However, most analyses focus on a limited number of models, domains, or feature sets. Consequently, it remains unclear **how broadly such linguistic signals generalize across different models and text domains**.

Recent work has also highlighted the importance of **robustness and generalization** in AI-text detection. Large-scale evaluations show that many detectors and linguistic signals degrade under distribution shift, such as unseen models, domains, or generation settings (Li et al., 2024a; Wang et al., 2024; Dugan et al., 2024). Moreover, some apparent linguistic cues may arise from confounding factors such as decoding strategies, prompts, or dataset artifacts rather than properties of machine-generated language (Li et al., 2024a; Reinhart et al., 2025). This raises a broader question: **to what extent reported linguistic cues reflect genuine properties of AI-generated language**.

3 Methods

In this work, we train a classifier based on interpretable linguistic features to distinguish between human-written and LLM-generated text. Using this classifier, we analyze which linguistic signals contribute most strongly to the prediction and evaluate their robustness across models and text domains. In this section, we describe the data used to train the classifiers, the linguistic features we extract, and the methodological steps taken to analyze linguistic signals of LLM-generated text.

3.1 Data

We use the MAGE dataset (Li et al., 2024a), a benchmark for detecting English LLM-generated text, where each instance pairs a human-written passage with continuations from 27 models from seven families, referred to as *model domains* henceforth (OpenAI GPT, LLaMA, GLM-130B, FLAN-T5, OPT, BigScience, and EleutherAI). The human-written texts come from 10 sources spanning multiple writing tasks: *opinion statements* (CMV, Yelp), *news articles* (XSum, TLDR), *question answering* (ELI5), *story generation* (WritingPrompts, ROCStories), *commonsense reasoning* (HellaSwag),

knowledge illustration (SQuAD), and *scientific writing* (SciGen) (size details in Section A.1). MAGE also provides a GPT-4-generated test set from four novel domains (CNN/DailyMail, DialogSum, IMDb, PubMed) to evaluate detection on completely out-of-distribution data.[1]

3.2 Linguistic Features

We extract *interpretable linguistic features* with the `elfen` Python package (Maurer, 2026) from various groups: surface-level, syntactic, and morphological structures as well as information-theoretic, lexical richness, semantic, and named entity features, and features based on measurements of emotional and psycholinguistic grounding of tokens. For surface-level, lexical richness, and readability features, we retrain only one measure from each set of theoretically equivalent features. For instance, both the type-token ratio (TTR) and the lemma-token ratio capture lexical richness, despite differing in formulation. Section A.3 provides a full overview of the features used in our experiments.

Except for raw counts of types, sentences, characters, lemmas, and syllables (e.g., the number of named entities in a text), we normalize count-based features by token count to obtain relative frequencies. We remove uniform features (i.e., features with only one value over the dataset) and handle missing and infinite values through feature-specific imputation: binary features are imputed with the mode, integer features with the median, and continuous features with either the mean (for normally distributed features) or median (for skewed distributions), ensuring that the statistical properties of each feature are preserved during preprocessing.

3.3 Classifier, Metrics and Evaluation

Classifier We use a linear Support Vector Machine (SVM) with class weighting to account for the strong class imbalance between human and LLM texts in MAGE (details on model choice and parameters in Section A.2).

Metrics and Evaluation We evaluate detection performance using three metrics: **Macro F1**, which accounts for class imbalance in test sets; **AUROC** (Area Under the Receiver Operating Characteristic curve) which measures discrimination between classes across decision thresholds; and **AvgRec** (Average Recall, details in Section A.2).

[1]Potential prompt effects are discussed via a cross-dataset experiment in Section A.1.

Testbeds We evaluate detection performance on eight testbeds from Li et al. (2024a) that vary in generalization scenarios (see summary in Table 1 and Section A.4 for details). While some testbeds operate at fine-grained model granularity (e.g., TB1 evaluates domain–model pairs across all 27 models), we present most analyses at the level of model families to reduce noise from individual model variation and improve interpretability.

3.4 Experiments

We run two sets of experiments: (1) LLM-authorship detection (**RQ1**), which evaluates the effectiveness and robustness of classifiers based on interpretable linguistic features, and (2) feature area ablation (**RQ2** and **RQ3**), which analyzes the contribution of different linguistic feature groups.

3.4.1 LLM-Authorship Detection

Classifiers are evaluated under **in-domain (ID)** and **out-of-domain (OOD)** conditions across three evaluation scenarios: general classification, model domain effects, and text domain effects. General classification uses **TB4** and **TB1** in ID settings, and **TB8** and **TB7** in OOD settings. Model domain effects are evaluated with **TB2** (ID) and **TB5** (OOD), while text domain effects are evaluated with **TB3** (ID) and **TB6** (OOD).

3.4.2 Feature Area Ablations

We conduct two types of ablation studies to investigate feature area importance: **(A)** Leave-one-out, where classifiers are trained on all eleven feature areas but one, and **(B)** Cumulative, where feature areas are removed sequentially according to their importance in (A), and classifiers are retrained with progressively fewer feature areas. We run (A) on all testbeds and (B) only for the domain-agnostic single-classifier settings (**TB4** and **TB7**) to avoid the combinatorial explosion of ablation classifiers.

4 Results

This section first presents the results of the LLM-authorship detection experiments addressing **RQ1** and then analyzes feature-area ablations to address **RQ2** and **RQ3**.

4.1 LLM-Authorship Detection

We first examine general detection performance, followed by an analysis of model and text domain effects. Table 2 presents the results, where our classifier appears as "SVM w/ Ling. Feats.", while the

Name	Setting	Description
TB1	Text domain & model specific	One classifier per text domain–model pair; trained and tested on data from the same domain–model combination.
TB2	Arbitrary text domains & model domain-specific	One classifier per model domain; trained and tested on all text domains but using data from only one specific model.
TB3	Fixed text domain & arbitrary models	One classifier per text domain; trained and tested on all model domains but using data from only one specific text domain.
TB4	Arbitrary text domains & arbitrary models	A single classifier trained and tested on all available data across all text and model domains.
TB5	Arbitrary text domains & unseen model domain	Leave-one-model-domain-out evaluation: trained on all data except one model domain and tested on the held-out model domain.
TB6	Unseen text domains & arbitrary models	Leave-one-text-domain-out evaluation: trained on all data except one text domain and tested on the held-out text domain.
TB7	Unseen text domains & single unseen model	Trained on all available data and evaluated on entirely new text domains (CNN, DialogSum, IMDb, PubMed) generated by GPT-4.
TB8	Unseen text domain & unseen model domain	Leave-one text–model pair out: trained on all data except one text–model domain pair and tested on the held-out pair; ; similar to TB7 but using existing domains to control for text–model pair effects.

Table 1: Evaluation testbeds used to analyze classifier robustness across text domains and model domains.

Setting	Method	AvgRec	AUROC	Macro F1
TB1	SVM w/ Ling. Feats.	.954	.987	.788
TB4	FastText	.788	.83	-
	GLTR	.554	.74	-
	Longformer	.905	.99	-
	SVM w/ Ling. Feats.	.840	.968	.827
TB7	FastText	.703	.74	-
	GLTR	.577	.73	-
	Longformer	.758	.94	-
	SVM w/ Ling. Feats.	.818	.907	.808
TB8	SVM w/ Ling. Feats.	.806	.945	.588

Table 2: Aggregate classification results: *In-domain* (**TB1** and **TB4**) and *Out-of-domain* (**TB7** and **TB8**); AvgRec and AUROC for comparison to MAGE baselines (FastText, GLTR and Longformer by Li et al. (2024a)) and Macro F1 for cross-experiment comparability.

Model Domain	ID (TB2)	OOD (TB5)	Difference (TB5 - TB2)
OPT	.914	.714	-.199
Eleuther	.881	.528	-.353
FLAN-T5	.766	.673	-.092
LLaMA	.753	.644	-.110
BigScience	.656	.595	-.061
GLM	.618	.496	-.122
OpenAI	.555	.616	+.061
Mean	.735	.609	-.125

Table 3: Macro F1 results of ID (**TB2**), OOD (**TB5**), and their difference per model domain. TB2 trains and tests on the same model domain data with arbitrary domains, while TB5 excludes the target model domain from training, ordered by ID performance (descending).

remaining methods come from Li et al. (2024a). In the mixed in-domain setting (TB4), where a single classifier is trained and tested on all data, our method achieves 82.7% F1. While this score indicates some room for improvement, the AUROC comparison shows that our simple and interpretable method surpasses three substantially more resource-demanding and blackbox methods by a large margin and is only 2% below the best-performing Transformer-based model. When the classifier is trained and tested separately on each text domain-model pair (TB1), the average performance decreases to 78.8% F1, suggesting domain effects on the classification performance. These domain effects are much more pronounced in the OOD experiments (TB8), where the average performance drops to 58.8% (23.9% performance drop from the ID setting in TB4). Finally, in the unseen text domain and model experiment (TB7), we achieve 80.84% F1 and only 3% lower AUROC compared to Li et al. (2024a). This set of unseen text domains-model pair seems far less challenging than some of the other text and model domain pairs, as evident in TB8. We now zoom into these domain effects on classifier performances.

Model Domain Effects Table 3 presents the classifier performances in ID and OOD experiments. When trained and tested on the data from the same model domain (ID), the classifiers perform differently for each model domain. For OPT and Eleuther, the classifiers perform well above the domain-agnostic ID experiments (by $+8.7\%$ and $+5.4\%$ resp.; cf. Table 2, TB4, 82.7% to Table 3, ID), and for BigScience, GLM, and OpenAI, well below (by -17.1%, -20.9%, and -27.2%). This indicates that certain model domains exhibit distinct linguistic signatures that alone provide sufficient signal for the model domain-specific classifiers. Looking at the OOD performance, where the performance drops by 19.9% and 35.3%, for OPT and Eleuther, respectively, it becomes clear that the presence of such linguistic model domain idiosyncracies in the training data is vital for detectability.

Task	Text Domain	ID (TB3)	OOD (TB6)	Difference (TB6 - TB3)
Story Generation	WP	.964	.883	-.081
	ROCT	.928	.668	-.260
Commonsense Reasoning	HSwag	.953	.739	-.214
News Articles	XSum	.941	.897	-.044
	TLDR	.908	.704	-.205
Scientific Writing	SciGen	.920	.927	+.008
Question Answering	ELI5	.873	.807	-.066
Knowledge Illustration	SQuAD	.829	.667	-.163
Opinion Statements	CMV	.924	.866	-.058
	Yelp	.804	.777	-.028
	Mean	.904	.794	-.111

Table 4: Macro F1 results of ID (**TB3**), OOD (**TB6**) and their difference. TB3 trains and tests on the same domain with arbitrary models, while TB6 excludes the target domain from training, ordered by ID performance.

Similarly, for the remaining model domains, we observe a moderate to low performance drop (by 12.2% to 6.1%). Surprisingly, in the OOD setting, the performance increases for the OpenAI models, suggesting that the models display linguistic markers that generalize over LLMs, which can be useful in OOD settings provided sufficient variety in LLM-generated training data.

Finally, we examine the unseen text–model domain setting (TB8). The results, summarized in Figure 1 (left), reveal similar patterns to TB5: classifiers performing worse for GLM, Eleuther, BigScience, and OpenAI (48.1%, 51.0%, 57.6%, and 59.7%), and better for OPT, FLAN, and LLaMA on average (68.4%, 63.7%, 62.8%).

Text Domain Effects Table 4 presents the classifier performances in both ID and OOD experiments. When trained and tested on the data from the same text domain (ID), discrepancy is less pronounced than in the model domain experiments, suggesting that text domain effects are not as detrimental to classifier performance as model domain for in-domain classification (cf. Table 3, ID to Table 4, ID). For most of the domains, the ID performance is above 90% F1, and the lowest performance is still relatively high, at 80.42%. Nonetheless, the OOD experiments reveal several noteworthy patterns: for ROCStories, HellaSwag, and TLDR, performance drops most sharply, by 26.0%, 21.4%, and 20.5%, respectively; this is consistent with their larger distributional differences (see Section E.1). For

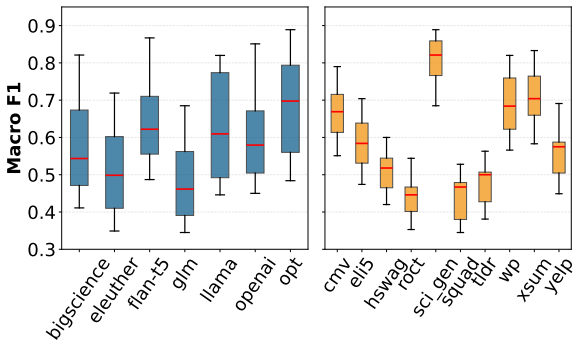


Figure 1: Distribution of Macro F1 scores across **TB8** (unseen text domain-model domain pairs), left pane for model domains and the right pane for text domains.

SQuAD, the drop is more moderate, at 16.3%; for SciGen, performance improves slightly, by 0.8%; and for the remaining domains, performance drops only marginally. Thus, the availability of text-domain data affects the classifier performance, and at times, quite substantially. Finally, for the unseen text and model domain setting (TB8; Figure 1, right), we observe a large performance discrepancy ranging from 71% to 43.7% F1[2].

Answer to RQ1: Linguistic features provide robust and generalizable signals for distinguishing LLM-generated from human-authored text across models and text domains. Feature-driven classifiers achieve comparable results to resource-intensive blackbox methods while relying on simple and interpretable representations. Our results indicate the presence of generalizable linguistic markers of LLM-generated text; however, differences in ID and OOD performance highlight substantial influences of model and text domain.

4.2 Features of AI-Authored Texts

We now zoom in on the effects of the types of linguistic features on the classification performance. We start by discussing the general classifiers and then focus on the model and text domains.

Table 5 displays the results from the leave-one-out feature area ablation on the ID domain-agnostic classifier (TB4) and the OOD setting (TB7). In the TB4 setting, we make two observations. First, dropping *lexical richness* features results in a stark performance drop of −13.1%, far larger than for any other feature area. The next most influential area is *information*, with −1.8% drop. Second, the effects

[2]Similarly for the individual text domains in TB7 with 96.82%, 87.18%, 83.10% and 50.24% F1 for CNN, pubmed, imdb and dialogsum, respectively (see Appendix, Table 11).

		Baseline	Surface	LexRich.	Emotion	Psycholing.	Read.	Morph.	POS	Depend.	Semantic	Entities	Inform.
F1 Macro	**TB4**	**.827**	.822	.696	.824	.821	.827	.824	.823	.823	.825	.827	.809
	TB7	**.808**	.818	.531	.814	.834	.806	.846	.815	.819	.808	.810	.824
# Feats. Removed		**0**	6	3	36	77	2	58	18	47	16	19	2

Table 5: Leave-one-out ablation results for **TB4** (arbitrary text domains & arbitrary models) and **TB7** (unseen text domains & model). Classifier performances when each feature area is dropped from the full feature set.

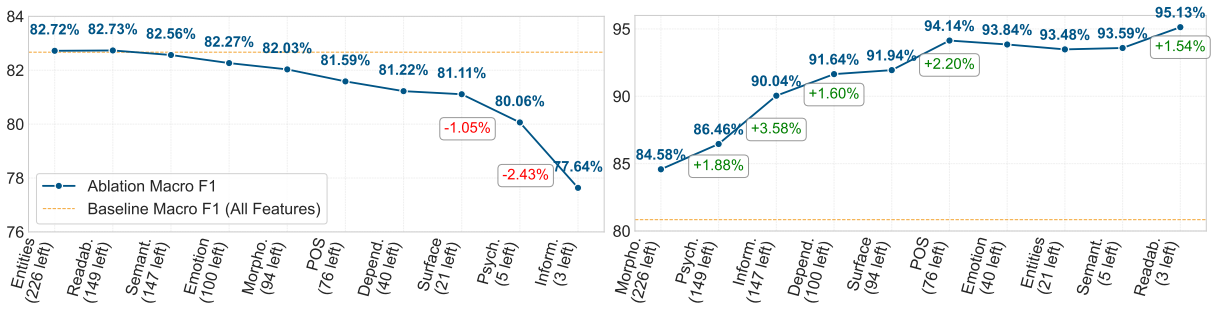


Figure 2: Macro F1 performance changes with cumulative feature area removal. The left pane shows results for **TB4**, and the right pane for **TB7**, in performance impact order from Table 5.

from the remaining 10 feature areas are marginal, with the largest decrease of -0.23%. These effects align with the cumulative ablation results (left pane of Figure 2). Dropping the *readability* and *entities* areas initially increases the performance. As the remaining areas are removed one by one, the performance slowly degrades. However, even after removing ten areas, the classifier retains much of its performance, dropping only about 5% from the full-feature baseline (from 82.67% to 77.64%).

In the OOD setting (TB7 in Table 5), the results for *lexical richness* are similar to the ID setting. For the remaining areas, the results vary, with a slight performance decrease for the *semantic* and *readability* areas (-0.06% and -0.28%) and an increase for the remaining ones (max. $+3.74\%$). Unlike the ID setting, the cumulative ablation here reveals an interesting trend (see right pane, Figure 2), where removing the first six feature areas improves performance by $+13.3\%$, and continuing with the next three areas degrades performance by -0.55%. Furthermore, in the presence of only the *lexical richness* and *readability* features, *readability* features similarly degrade the performance, with an increase in F1 by $+1.54\%$ when removed. To understand this trend, we now discuss a similar ablation for TB8, the data for which we can compare the results across experiments.

As a complete ablation of 70 pairs (770 classifiers) is not feasible, we run the leave-one-out ablation on four text domains and four model domains based on performance analysis (see Section C.3), in total 16 pairs. We display the findings in Figure 6. The experiments reveal two major findings: 1) *lexical richness* features systematically characterize AI-authored texts (except for the SQuAD text domain) with a varying degree, 2) other feature areas like *surface*, *psycholinguistic*, *information*, *dependency*, and *morphological* display varying levels of informativeness depending on text domain and model domain pairs. For XSum with LLaMA or OpenAI pairs, the performance increases with the removal of all four feature areas but *information* (detailed discussion in Section C.4).

Model Domain Effects Figure 3 presents the feature area ablation results under the ID setting. Two trends are immediately apparent. First, removing the *lexical richness* features from training yields opposing effects across models: performance improves for OpenAI, LLaMA, and GLM (+0.31%, +2.20%, and +2.54%, respectively), whereas it decreases for FLAN-T5, OPT, BigScience, and Eleuther (9.5%, 3.23%, 5.71%, and 2.44%, respectively). Second, for OpenAI and LLaMA models, the classifier performs better without several feature areas, i.e., *lexical richness*, *surface*, and *readability* all slightly improve F1 when removed. Conversely, simpler models FLAN-T5 and BigScience show consistent and larger drops across most area removals, indicating more salient linguistic signatures. The ablation in the OOD setting

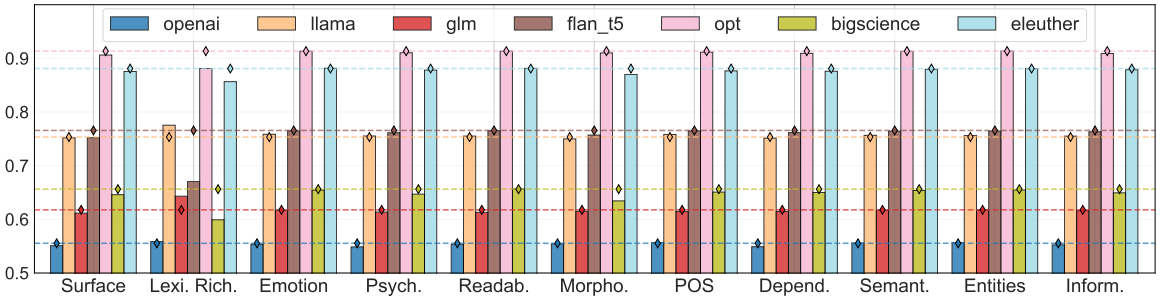


Figure 3: Ablation results on **TB2**. The horizontal dashed lines indicate the baseline results (all features) and the bars indicate the performance change in Macro F1 after removing the corresponding feature area.

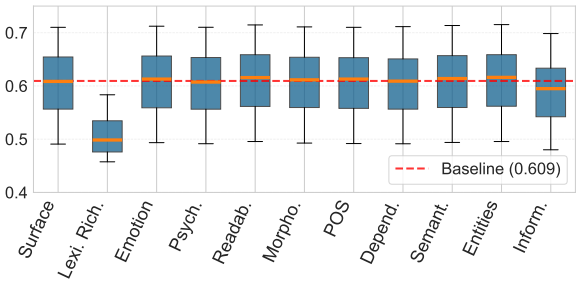


Figure 4: Ablation performance ranges on **TB5** across the 11 feature areas after removing the corresponding feature area compared to the baseline (dashed red line).

shows interesting patterns (see Figure 4). Dropping *lexical richness* results in a stark performance drop for all models this time, followed by *information* features. This suggests *lexical richness* and *information* features encode salient signals that generalize across models and are critical for OOD detection, whereas in the ID setting, the classifier can likely learn sufficient patterns from the remaining features. To provide qualitative intuition for these patterns, Section D presents representative human- and LLM-generated examples illustrating differences in the most influential feature areas.

Answer to RQ2: *Lexical richness* has the largest impact on classifier performance, emerging as the strongest indicator of LLM-generated text. Feature ablations on the model domains further indicate that models cluster into two groups: a) OpenAI, Llama, and GLM, and b) FLAN-T5, OPT, and BigScience. One possible explanation is that models within each group share similarities in architectures, training data, or instruction-tuning, which leads to similarities in their linguistic markers. To further investigate this observation, we analyze correlations of feature-area contributions across model families (Section E.2), which provide statistical support for these model groupings.

4.3 Text Domain Effects

Figure 5 displays the feature area ablation results on the ID and OOD settings. For the ID setting (upper pane), dropping *lexical richness* consistently degrades the classifier performance across all ten text domains (ranging from 2.5% for HellaSwag to 11.8% for ELI5) but one, SQuAD (+0.19%), which is intuitive as SQuAD answers are based on a provided context and are typically short phrases unlikely to vary considerably in lexical richness. For this domain, the areas that exhibit the largest performance drops are *surface*, *emotion*, and *POS*, again intuitive as different choices of answer spans would result in variance in these features. Second, the effects of the remaining feature areas on the text domains vary quite a lot, without a clear trend for types of texts such as opinion statements or story generation. Similar to observations from the model domain experiments, in the OOD setting (lower pane in Figure 5), *lexical richness* features carry crucial information, with drops ranging from 27.45% (XSum) to 0.3% (SQuAD)[3]. Additionally, observed from Figure 6, for XSum, *readability*, *psycholinguistics*, *morphological*, and *surface* (also for WritingPrompts) features all seem noisy when both text and model domain pair is unseen.

Answer to RQ3: Feature areas display varying levels of importance in text domains. While lexical richness[4], given its idiosyncrasy across LLMs, remains the most influential feature area, the importance of other areas varies substantially. To better understand these differences, we analyze similarities between text domains by comparing feature distributions using Wasserstein distances (Section E.1). The results reveal differences between domains that help explain variation in feature

[3]Similar effects can be observed for TB7 (see Section C.2, Figure 9.)

[4]Further analyses on this feature area in Section C.4.

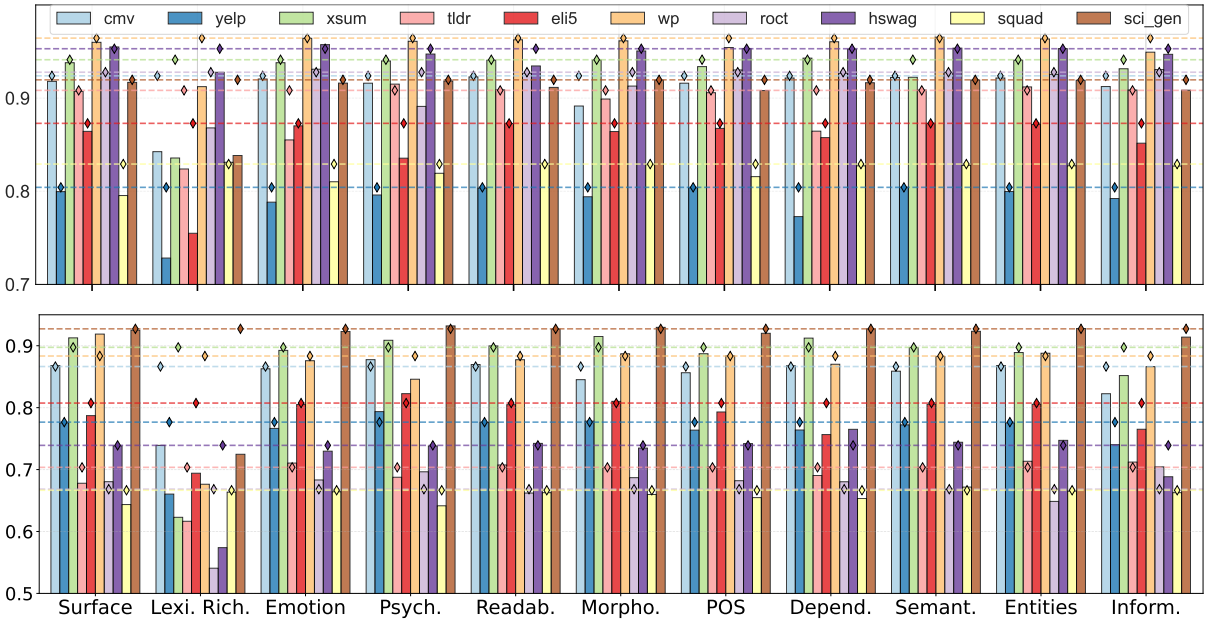


Figure 5: Macro F1 results of ablation study, top pane: **TB3** and bottom pane: **TB6**. The horizontal dashed lines indicate baseline results for each model, and the bars the performance after dropping the corresponding feature area.

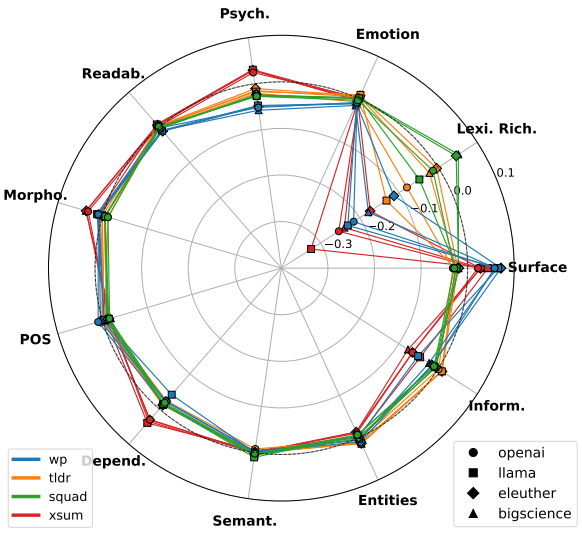


Figure 6: Ablation performance variance across 16 text domain-model domain after removing the corresponding feature area (selection based on Section C.3).

informativeness and classifier performance.

5 Conclusions and Discussion

LLM-generated text increasingly populates both formal and informal discourse. Understanding the linguistic markers that signal such text is therefore essential, both to develop effective detection methods and to provide interpretable explanations to participants of such discourse.

Through extensive classification experiments, we showed that linguistic features (interpretable to both experts and non-experts) are highly informative of whether a text is generated by an LLM. At the same time, the linguistic markers of generated text are far from uniform. Many indicators previously proposed in the literature turn out to depend strongly on particular models or text domains. In contrast, measures of lexical richness consistently emerge as robust signals across contexts. Beyond individual features, systematic patterns also appear across groups of LLMs and types of text domains, suggesting that both model architecture and discourse context shape the characteristics of generated language.

These findings highlight both the promise and the limitations of linguistically grounded explanations of AI-generated text. While some signals generalize across contexts, others depend on specific models or domains. As a result, detection systems deployed in real-world settings, such as recognizing LLM-generated peer reviews or papers, must be evaluated under diverse conditions and updated as new models emerge. More broadly, our results emphasize the importance of systematically assessing the robustness of linguistic signals when using them to interpret or detect AI-generated language.

Limitations

Our setup and findings are limited in several ways. Firstly, while we find particular patterns in English data, this may not hold for other languages. In principle, our methods could be easily applied to any language that (a) has the necessary benchmarks with a broad coverage available, and (b) external tooling to extract the features. The feature extraction tool we use covers a large number of languages, but has some blind spots, particularly regarding low-resource languages and/or languages from the African continent. Moreover, the applicability of the feature definitions, especially for lexical richness, depends on the particular language. Results in other languages may thus not necessarily be strictly comparable.

Secondly, while we use, to the best of our knowledge, the broadest and most current available benchmark, it does not cover the newest models. Our results thus have limited informativeness on the behavior of these models. While this is outside of the scope of this paper, it underlines the need for continually updated benchmarks to assess whether findings on one set of models still hold.

Finally, our experiments cover two important dimensions that impact the detectability of machine-generated text: text domain and model (family). We do not consider prompt variants as a potential source of performance variation. Though outside of the scope of this paper and challenging due to a lack of availability of resources covering all three dimensions, we provide a small pilot experiment including prompt formulation as a factor of performance variation on one of the datasets (CMV) in Section B.2.

Ethical Considerations

While our work advances understanding of the features and quality of LLM-generated texts, it raises several important ethical considerations. First, insights from our findings raise potential dual-use concerns. On one hand, they could be misused by malicious actors to develop tools that better mimic human writing patterns. It is therefore essential not only to highlight these risks but also to support community efforts to identify, mitigate, and safeguard against such misuse. On the other hand, understanding the linguistic signatures of AI-generated text has constructive applications. These insights can help guide generation toward more appropriate content, for example, by suggesting reformulations of user texts to match a desired stylistic profile. Additionally, classifiers based on fully interpretable linguistic features can serve as effective educational tools, helping the public recognize LLM-generated content. Finally, all human-authored texts used in our analyses were drawn from publicly available datasets and handled in accordance with established ethical research standards. No identifiable or private user data was used. Nonetheless, ongoing reflection on issues of consent, data provenance, and user agency remains vital when working with human discourse.

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A Methods

A.1 Dataset Details

MAGE Dataset We use the MAGE dataset (Li et al., 2024a), comprising human-written texts from 10 text domains and corresponding AI-generated continuations from 27 models across 7 model domains (families). The 10 text domains span diverse writing tasks: opinion statements (CMV, Yelp), news articles (XSum, TLDR), question answering (ELI5), story generation (WritingPrompts, ROCStories), commonsense reasoning (HellaSwag), knowledge illustration (SQuAD), and scientific writing (SciGen). The 7 model families (domains) are OpenAI (GPT-3.5-turbo, text-davinci-002/003), LLaMA (7B, 13B, 30B, 65B), GLM-130B, FLAN-T5 (small, base, large, xl, xxl), OPT (125M to 30B, including IML variants), BigScience (BLOOM-7B, T0-3B, T0-11B), and EleutherAI (GPT-J-6B, GPT-NeoX-20B).

Machine-generated texts were created using three prompt types: continuation prompts (first 30 words of human text), topical prompts (based on titles/topics), and specified prompts (topical prompts with source information). We only used continuation prompts to avoid confounding effects from prompt variation. Table 6 shows the data distribution across text domains and splits.

CMV Dataset To assess prompt generalization, we conducted cross-dataset experiments using the ChangeMyView (CMV) dataset from Dönmez and Falenska (2025), which employs direct response prompts rather than continuation prompts. We evaluated: (1) training on MAGE-CMV and testing on their CMV dataset, and (2) training on their dataset and testing on MAGE-CMV. This tests whether models can generalize across different prompting strategies for the same domain. Statistics of the final CMV dataset are in Table 7.

A.2 Classifier, Metrics, and Evaluation

We chose a linear Support Vector Machine (SVM) for its effectiveness in high-dimensional feature spaces (284 features), computational efficiency for training hundreds of classifiers across testbeds, and interpretability through feature coefficients.

Implementation We employed a linear Support Vector Machine (SVM) using scikit-learn's `SVC` (v1.7.0 [5]) implementation with the following configuration: linear kernel, regularization parameter ($C = 1.0$; default value), balanced class weighting to handle class imbalance, random seed 42 for reproducibility. The training was accelerated using Intel Extension for Scikit-learn (`sklearnex`).

Features were standardized using z-score normalization (zero mean, unit variance) via `StandardScaler`, fitted on the training set and applied to validation and test sets.

[5] https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html

Split	CMV	Yelp	XSum	TLDR	ELI5
Train	4,223H / 16,556AI	31,827H / 20,529AI	4,708H / 21,255AI	2,826H / 16,092AI	16,706H / 20,764AI
Val	2,436H / 2,023AI	2,657H / 2,564AI	3,274H / 2,655AI	2,526H / 1,971AI	3,146H / 2,566AI
Test	2,403H / 2,039AI	2,652H / 2,547AI	3,283H / 2,654AI	2,535H / 1,981AI	3,156H / 2,597AI
Split	**WP**	**ROCT**	**HSwag**	**SQuAD**	**SciGen**
Train	6,356H / 20,005AI	3,287H / 20,712AI	3,129H / 20,295AI	15,820H / 19,940AI	4,436H / 18,691AI
Val	3,133H / 2,481AI	3,284H / 2,583AI	3,288H / 2,521AI	2,524H / 2,475AI	2,531H / 2,312AI
Test	3,099H / 2,538AI	3,275H / 2,588AI	3,292H / 2,535AI	2,508H / 2,496AI	2,538H / 2,251AI
Overall Total: Train (93,318H / 194,839AI), Val (28,799H / 24,151AI), Test (28,741H / 24,226AI)					

Table 6: MAGE dataset statistics using continuation prompts only. Training includes all available AI generations from 27 models across 7 families (model domains); validation and test sets are approximately balanced (H = Human, AI = AI-generated).

Source	Number of Samples
Human	157,880 comments (13,498 posts)
GPT	13,498 (1 comment per post)
LLaMA	13,489 (1 comment per post)
Mistral	13,498 (1 comment per post)

Table 7: Final dataset sizes for the ChangeMyView (CMV) corpus used in training and evaluation.

Evaluation Metrics Model performance was assessed using three primary metrics—**Macro F1-score:** Selected as the primary metric due to possible class imbalance in test sets. Unlike micro-averaged metrics or accuracy, Macro F1 treats both classes equally by averaging their individual F1 scores, **AUROC (Area Under ROC Curve):** Measures the classifier's ability to discriminate between human and AI text across all possible decision thresholds, providing a threshold-independent evaluation, **Average Recall (AvgRec):** The arithmetic mean of recall scores for both classes, calculated as $\text{AvgRec} = \frac{\text{Recall}_{\text{human}} + \text{Recall}_{\text{AI}}}{2}$. It ensures balanced evaluation regardless of class imbalance.

Accuracy was also reported for completeness. Additional metrics included per-class precision, recall, and F1-scores for both human and AI classes.

Reproducibility Resources All experiments were conducted using Python 3.10.18 with the following key dependencies: scikit-learn 1.7.0[6] for SVM implementation and evaluation metrics, Intel Extension for Scikit-learn (sklearnex) 2025.9.0[7] for accelerated training, pandas 2.3.0[8] for data manipulation, numpy 2.2.6[9] for numerical operations, matplotlib 3.10.3[10] and seaborn 0.13.2[11] for visualization, spaCy 3.8.7[12] with model en_core_web_lg 3.8.0, and ELFEN 1.1.9[13] for linguistic feature extraction. More details and required packages can be found in the requirements file (*requirements.txt*) on the repository. All experiments used consistent random seeds (seed=42) for train-test splits and model initialization.

A.3 Full List of Linguistic Features

In the following, we provide an overview of the full set of features used in our experiments. We report features per feature area and rationales for the selection of features wherever we do not use all the available features.

A.3.1 Surface-level

We use the number of tokens, characters, sentences, lemmas, and words over 5 characters, and the raw sequence length in characters (including whitespaces and special characters). We do not consider all additional surface-level features, as we consider them theoretically equivalent to the ones we select.

We do not include the number of types (unique tokens), as it is expected to carry similar information as the number of tokens and the number of lemmas. We do not include sentence-normalized measures, as all information is captured by the combination of the number of sentences and the respective measure to be sentence-normalized. We do not include the average word length, as similar but more interpretable information is captured by the number of long words.

[6] https://scikit-learn.org
[7] https://uxlfoundation.github.io/scikit-learn-intelex/
[8] https://pandas.pydata.org
[9] https://numpy.org
[10] https://matplotlib.org
[11] https://seaborn.pydata.org
[12] https://spacy.io
[13] https://github.com/mmmaurer/elfen

A.3.2 Parts-of-Speech

We use the number of tokens per universal dependencies POS tag (de Marneffe et al., 2021), and the POS variability (i.e., the number of different POS tags divided by the number of tokens).

A.3.3 Readability

We use the total number of syllables in a text, and the Flesch reading ease (Kincaid et al., 1975) as measures of readability. We do not consider all other readability features, as we consider them theoretically equivalent to the Flesch reading ease or the number of syllables. More specifically, they are all designed to measure the reading level, either measured in school grade levels, age, or a more abstract index of how easy to read a given text is.

A.3.4 Lexical Richness

We use the type-token ratio (TTR), the number of global token hapax legomena (i.e., the number of tokens per text that only occur once across the whole dataset), and the lexical density (i.e., the percentage of adjectives, nouns, verbs, and adverbs per text). We do not consider all other lexical richness features, as we consider them theoretically equivalent to at least one of the selected ones. More specifically, they all measure how many different *kinds* of words relative to the overall number of words are present in a text. Our choice of TTR gives us a general measure, the global token hapax legomena provides information about rare words, and the lexical density about content words.

A.3.5 Information-theoretic

We use both information-theoretic features, the compressibility and Shannon entropy of a text.

A.3.6 Named Entities

We use the number of named entities per entity type and the number of entities overall in a text.

A.3.7 Semantic

We use the number of hedges in a text, the average number of synsets overall and per nouns, verbs, adjectives, and adverbs, and the number of tokens with a high and with a low number of synsets overall and per nouns, verbs, adjectives, and adverbs, respectively.

A.3.8 Morphological

We use all available universal dependencies morphological features that are not uniform in our data.

A.3.9 Syntactic Dependencies

We use the number of universal dependency relations per relation type, and the dependency tree width, tree depth, tree branching factor, the ramification factor, and the number of noun chunks.

A.3.10 Emotion and Sentiment

We use all emotion and sentiment features available in `elfen`. This includes the number of positive and negative sentiment tokens per text, the number of tokens with a high and a low valence, arousal, and dominance, the average valence, arousal, and dominance, the average intensity, and the number of high and low intensity tokens per Plutchik emotion.

A.3.11 Psycholinguistic

We use all available psycholinguistic features. This includes the number of tokens with high and low concreteness, age of acquisition, prevalence, socialness, iconicity, and sensorimotor association, and the average rating across tokens per text for these dimensions, respecively.

A.4 Testbeds

We built on the settings designed by Li et al. (2024a) and created a similar setup of eight primary testbeds to evaluate detection performance across various generalization scenarios:

TB1 (Fixed-domain & Model-specific) trained and tested classifiers on individual domain-model pairs, establishing baseline performance for 10 text domains and 27 AI models (270 classifiers). **TB1-1** extended this setting by grouping models into model families/domains (OpenAI, LLaMA, GLM, Flan-T5, OPT, BigScience, Eleuther) to assess model domain level of detection within fixed text domains and model variant vs. model family/domain performance in fixed scenarios, resulting in 70 classifiers.

TB2 (Arbitrary-text-domains & Model-domain-specific) trained on all text domains for each model family/domain separately (i.e., trained on all domains, but only using AI text generated by one model family; i.e., in one model domain).

TB3 (Fixed-text-domain & Arbitrary-model-domains) evaluated cross-model generalization within individual text domains by training on all available models for each text domain separately.

TB4 (Arbitrary-text-domains & Arbitrary-model-domains) represented our most comprehensive training scenario, combining all available domains and models to assess overall detection capability.

TB5 (Unseen model domains) evaluated a scenario of leave-one-out by leaving data generated by a specific model family (model domain) out of the training and then evaluated on the excluded model domain to assess the generalization.

TB6 (Unseen text domains) similarly assessed text domain generalization by training on all text domains except one and testing on the excluded one.

TB7 (Unseen text-domains & Unseen-model) presented the most challenging scenario, testing on completely unseen text domains (CNN/DailyMail, DialogSum, IMDb, PubMed) generated by GPT-4, which was excluded from all training data. **TB7-1** extended this analysis by evaluating each unseen text domain individually to examine text domain-specific performance variations.

TB8 (Unseen text-model-domain Pairs) introduced a novel cross-generalization setting where both the target text and model domain were completely excluded from training, creating 70 unique held-out combinations (10 domains $\times$ 7 families) to assess the classifier's ability to generalize to entirely novel domain-model pairings. This was intended to validate more the scenario of TB7.

We provide details of the configuration for each individual setting in Table 8.

B Classification Results

B.1 Detailed MAGE Testbeds Results

Complete results for all MAGE testbeds are presented in Table 9. Main findings are discussed in section 4. Here we highlight more details of some testbeds.

TB1.1: Fixed domain & model domain combinations Table 10 presents results for 70 domain-specific, model-domain-specific classifiers. OPT achieves the highest average detectability (94.4% F1), followed by Eleuther (92.4% F1) and FLAN-T5 (89.1% F1). OpenAI and GLM are least detectable (71.6% and 74.9% F1). Across text domains, WritingPrompts shows highest detection rates (93.9% avg F1), while Yelp is most challenging (73.7% avg F1). Performance ranges from 56.3% (OpenAI on SQuAD) to 98.6% (FLAN-T5 on WritingPrompts).

TB7: Per-domain analysis Table 11 presents the performance across the four unseen domains generated by GPT-4. First, we evaluate the performance on the combined text from the four text domains, then we test on individual ones. The combined evaluation represents the average performance, but individual settings give more insight. CNN/DailyMail news articles achieve the highest detection rate (96.82% F1), while DialogSum proves most challenging (50.24% F1). PubMed (87.18% F1) and IMDb (83.10% F1) fall in between. These results align with the other OOD settings: News articles (CNN) show similar high performance to other news data (XSum), similar -to an extent- behavior of the QAs (PubMed) and review/opinion (IMDb) domains, but conversational text (DialogSum) proves that such domain has salient distinct linguistic features that cannot be detected easily from features fitted on other domains. We discuss further the results of unseen domains in the following paragraph.

TB8: Domain-model pair combinations Table 12 presents all 70 text domain-model domain pair results in the wilder OOD setting. As a more robust test setting to complement TB7 we find remarkable shifts in results across text and model domains. Performance varies considerably (34.5%–88.9% F1). OPT is most detectable across text domains (68.4% avg F1), while GLM and Eleuther are least detectable (48.1% and 51.0% avg F1). SciGen is the easiest text domain to detect across models (80.6% avg F1), while ROC-Stories and SQuAD are most challenging (44.0% and 43.7% avg F1).

B.2 ChangeMyView Cross-Dataset Experiments

To assess the impact of different prompt formulations, we run a small experiment on one of the datasets. For this, we choose and compare the performance of the CMV subset in MAGE (in this section referred to as MAGE-CMV and CMV subset from Dönmez et al. (2025) (in this subsection referred to as CMV).

In-Distribution Performance The classifier achieves almost perfect performance on ChangeMyView (CMV) data in all in-distribution settings, with Macro F1 scores exceeding 98.5%. Table 13

TB#	Name	Training	Testing	# Classifiers
TB1	Fixed-domain & Model-specific	Single text domain + Single model	Same text domain + Same model	270
TB1-1	Fixed-text-domain & Model-domain-specific	Single text domain + Model domain	Same domain + Same model domain	70
TB2	Arbitrary-text-domains & Model-domain-specific	All text domains + Model domain	All text domains + Same model domain	7
TB3	Fixed-text-domain & Arbitrary-model-domains	Single text domain + All models	Same text domain + All models	10
TB4	Arbitrary-text-domains & Arbitrary-model-domains	All text domains + All models	All text domains + All models	1
TB5	Unseen model domains	All text domains + 6 model domains	All domains + 1 held-out model domain	7
TB6	Unseen text domains	9 text domains + All models	1 held-out text domain + All models	10
TB7	Unseen-text & -model domains	All text domains + All models	4 new text domains + GPT-4	1
TB7-1	Unseen-text-domain & Unseen-model	All text domains + All models	1 of 4 new text domains + GPT-4	4
TB8	Unseen text-model-domain pairs	9 text domains + 6 model domains	1 held-out text domain + 1 held-out model domain	70

Table 8: Overview of experimental testbeds for AI-generated text detection. Each testbed evaluates different generalization scenarios, ranging from fixed conditions (TB1) to completely unseen domain-model combinations (TB7, TB8).

shows the detailed results.

Cross-Dataset Generalization We evaluate two cross-dataset scenarios: (i) training on our CMV dataset and testing on MAGE-CMV, and (ii) training on MAGE-CMV and testing on our CMV dataset. Both scenarios reveal a critical finding: **model domain alignment between training and test data dramatically impacts detection performance, while prompt formulation appears to have a smaller effect**.

Scenario 1: CMV → MAGE-CMV. When testing on MAGE-CMV using all available model domains, performance drops significantly (Macro F1: 76.11%, AI-Recall: 56.99%). However, when we restrict both training and test data to only the shared model domains (OpenAI and LLaMA), performance improves substantially to 86.93% Macro F1 and 81.93% AI-Recall. This indicates that including Mistral in training and other model domains (5 additional domains) in testing introduces a distribution mismatch that harms detection (as we can see in Table 14). This could still be caused by the prompt strategies, which leads us to reverse the setting and analyze the results.

Scenario 2: MAGE-CMV → CMV. The pattern is even more pronounced in the reverse direction. Training on all 27 models across 7 model domains from MAGE-CMV yields poor performance on CMV test set (Macro F1: 43.53%, AI-Recall: 23.37%). Focusing only on shared model domains (OpenAI and LLaMA) dramatically improves results to 84.95% Macro F1 and 92.44% AI-Recall—nearly doubling the F1 score and quadrupling the AI-Recall. Table 15 shows the complete results.

Key Findings: Cross-dataset detection succeeds when model domains align between training and test distributions. Training on diverse model domains without representation in the test set introduces noise that severely degrades performance. The less severe drops compared to the in-domain performance for the shared model domains indicate that, while having an impact, the prompt formulation seems to have less of an impact on detectability than other factors.

C Ablation Results

In this section, we discuss further the ablation studies conducted on all testbeds (ablation study **A**)

Setting	Method	Acc	AUROC	F1-Macro
In-distribution Detection				
1. Fixed-Domain & Model (270 classifiers)	SVM w/ Ling. Feats.	0.9424	0.9865	0.7880
1.1. Fixed-Domain & Model-domain (70 classifiers)	SVM w/ Ling. Feats.	0.9234	0.9839	0.8403
2. Arbitrary-Domains & Fixed-Model (7 classifiers)	SVM w/ Ling. Feats.	0.8443	0.9856	0.7347
3. Fixed-Domain & Arbitrary-Models (10 classifiers)	SVM w/ Ling. Feats.	0.9056	0.9860	0.9044
4. Arbitrary-Domains & Arbitrary-Models (1 classifier)	SVM w/ Ling. Feats.	0.8277	0.9682	0.8267
Out-of-distribution Detection				
5. Unseen Models (7 classifiers)	SVM w/ Ling. Feats.	0.7291	0.9521	0.6094
6. Unseen Domains (10 classifiers)	SVM w/ Ling. Feats.	0.7995	0.9625	0.7936
7. Unseen Domains & Unseen Model (1 classifier)	SVM w/ Ling. Feats.	0.8223	0.9066	0.8084
7.1. Unseen Domains & Unseen Model (per domain) (4 classifiers)	SVM w/ Ling. Feats.	0.8127	0.9422	0.7934
8. Unseen Domain-Model Pairs (70 classifiers)	SVM w/ Ling. Feats.	0.6864	0.9450	0.5875

Table 9: Summary results for all testbeds (TB1-8). For testbeds with multiple classifiers, metrics represent the mean performance averaged across all individual classifiers within that testbed. Results are grouped into *in-distribution* (TB1-4) and *out-of-distribution* (TB5-8) detection scenarios.

Model domain	CMV	ELI5	HSwag	ROCT	SciGen	SQuAD	TLDR	WP	XSum	Yelp	Avg.
BigScience	0.844	0.750	0.784	0.828	0.867	0.701	0.759	0.949	0.893	0.753	**0.813**
Eleuther	0.935	0.934	0.870	0.959	0.945	0.956	0.871	0.971	0.951	0.851	**0.924**
FLAN-T5	0.926	0.901	0.819	0.835	0.951	0.855	0.881	0.986	0.967	0.793	**0.891**
GLM	0.856	0.683	0.836	0.817	0.708	0.692	0.642	0.914	0.758	0.586	**0.749**
LLaMA	0.891	0.798	0.945	0.986	0.783	0.789	0.888	0.935	0.812	0.708	**0.854**
OpenAI	0.742	0.632	0.919	0.733	0.742	0.563	0.656	0.839	0.725	0.608	**0.716**
OPT	0.960	0.911	0.951	0.964	0.967	0.936	0.945	0.978	0.966	0.858	**0.944**
Avg.	**0.879**	**0.801**	**0.875**	**0.875**	**0.852**	**0.785**	**0.806**	**0.939**	**0.867**	**0.737**	**0.842**

Table 10: **TB1.1** results: F1-Macro performance for fixed domain and model domain combinations (70 classifiers total). Each cell represents a separate classifier trained and tested on data from one domain and one model domain. Row averages show model domain detectability; column averages show domain-specific performance.

Dataset	Accuracy	AUROC	F1 Macro
CNN	0.9717	0.9973	0.9682
DialogSum	0.5191	0.8206	0.5024
IMDb	0.8650	0.9679	0.8310
PubMed	0.8950	0.9828	0.8718
ALL FOUR	0.8223	0.9066	0.8084

Table 11: Results for **TB7**: unseen-domains and unseen-models, using the full set of features. The unseen model is GPT-4

and on domain-agnostic settings (ablation study **B** -cumulative).

C.1 Model and Text Domain Effects Additional Analysis

Table 16 presents complete ablation results for testbeds 2, 3, 5, and 6, examining feature importance across model domains and text domains. Figures 7 and 8 visualize these patterns.

Model domain Effects (TB2 & TB5) When training on arbitrary text domains with model-domain-specific data (TB2 ID, TB5 OOD), lexical richness consistently dominates across all model domains. However, impact magnitude varies substantially: removing lexical richness hurts FLAN-T5 most severely (-9.5% TB2, -14.1% TB5), while OpenAI models show minimal dependency (+0.3% TB2, -13.3% TB5). This suggests OpenAI generations rely less on distinctive vocabulary patterns, making them harder to detect through lexical features alone. Surface features show consistent small negative effects (-0.4% to -1.4%), while most other features have negligible impact.

Text Domain Effects (TB3 & TB6) Domain-specific patterns (TB3 ID, TB6 OOD) reveal greater variation than model effects. Figure 8 illustrates these differences through performance distributions across feature ablations. In TB3 (ID, baseline 0.904), lexical richness shows the largest drop (median 0.85) but with tight distribution, while other features remain close to baseline with minimal variance. This indicates consistent, predictable behavior when text domains are seen during train-

Model domain	CMV	ELI5	HSwag	ROCT	SciGen	SQuAD	TLDR	WP	XSum	Yelp	Avg.
BigScience	0.642	0.557	0.530	0.433	0.821	0.411	0.456	0.684	0.704	0.517	**0.576**
Eleuther	0.585	0.505	0.443	0.370	0.719	0.349	0.399	0.608	0.636	0.492	**0.510**
FLAN-T5	0.669	0.644	0.600	0.544	0.867	0.487	0.507	0.724	0.739	0.589	**0.637**
GLM	0.551	0.474	0.420	0.353	0.685	0.345	0.381	0.566	0.583	0.449	**0.481**
LLaMA	0.725	0.633	0.487	0.446	0.813	0.471	0.507	0.820	0.790	0.586	**0.628**
OpenAI	0.706	0.584	0.518	0.450	0.851	0.467	0.500	0.636	0.683	0.575	**0.597**
OPT	0.790	0.704	0.559	0.484	0.889	0.528	0.563	0.795	0.833	0.691	**0.684**
Avg.	**0.667**	**0.586**	**0.508**	**0.440**	**0.806**	**0.437**	**0.473**	**0.691**	**0.710**	**0.557**	**0.587**

Table 12: Results of **TB8**; Macro F1 performance on unseen domain-model-domain pairs. Training data excludes both the target domain and target model domain, creating 70 unique domain-model combinations. Each cell shows the F1-Macro score when testing on a domain-model pair that was completely excluded from training. The **Avg** row shows the average performance across all text domains for each model domain, while the **Avg** column shows the average performance across all model domains for each domain. Overall statistics: Mean = 0.587, Std = 0.140, Range = [0.345, 0.889].

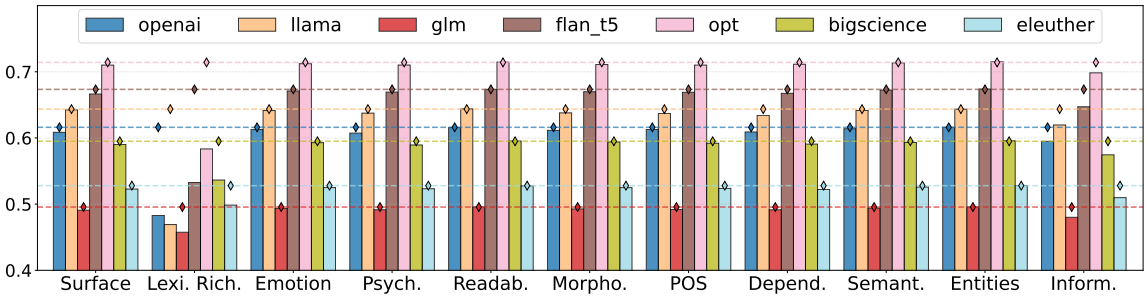


Figure 7: Visualization of results of ablation study on **TB5** (unseen models). The horizontal dashed lines indicate the original baseline results for each model, and the bars indicate the results of dropping the corresponding feature area.

AI Text	Accuracy	AUROC	F1 Macro
AI-combined	0.9884	0.9973	0.9884
GPT	0.9911	0.9994	0.9911
Llama	0.9852	0.9948	0.9852
Mistral	0.9920	0.9986	0.9920

Table 13: Results for **CMV dataset** for the different settings using the complete features set (284 features). *The use of accuracy here is due to the balanced dataset*

ing.

TB6 (OOD, baseline 0.794) shows increased variance across nearly all features. Lexical richness maintains strong impact (median 0.66) but with much wider spread, reflecting domain-dependent sensitivity: XSum shows massive drops in OOD (-27.4%), while SQuAD remains stable (-0.3% ID, -0.3% OOD). Surface features shift upward in TB6, often exceeding baseline for some text domains (XSum +1.5%, WP +3.5%), contrasting with their negative effects in TB3. However, their large variance (Figure 8b) proves how their behavior changes across different domains in the OOD setting compared to ID. Psycholinguistic features help formal domains when unseen (CMV +1.1%, XSum +1.1%, SciGen +0.5%) but hurt creative writing (WP -3.7%, ROC -3.7%). Information features consistently hurt OOD performance across most domains (-4.4% to -5.1%), suggesting they capture domain-specific patterns that don't generalize. Both psycholinguistic and information features display wider distributions in TB6, with outliers in both directions indicating divergent effects

C.2 Zoom-in TB7: Unseen Domains and Unseen Model

For the setting of completely unseen text and model domains, we tested two configurations: combining all four unseen domains (CNN, PubMed, IMDb, DialogSum) into a single test set, and testing each domain separately.

Combined Unseen Domains With all four domains combined (as seen in Table 5 in sec. 4.2), lexical richness removal causes the sharpest drop

Mage Test data	Accuracy	AUROC	F1 Macro	AI-Recall	CMV-Training size	MAGE-CMV-test size
i- All models	0.7765	0.7635	0.7611	0.5699	28,339 H, 28,340 AI	2,403H, 2,039 AI
ii-OpenAI-Llama	0.9191	0.9410	0.8693	0.8109	18,890 H, 18,891 AI	2,403 H, 550 AI

Table 14: Results of two settings for cross-CMV-MAGE setup: (i) classifier is trained on training split from ChangeMyView dataset with all AI-texts (GPT, LLaMA, & Mistral) and tested on MAGE-CMV data with AI-texts from all 27 models. (ii) classifier trained on training split from ChangeMyView but with AI-texts from only GPT & LLaMA, then tested on MAGE-CMV with AI-texts from OpenAI & LLaMA models.

Mage Test data	Accuracy	AUROC	F1 Macro	AI-Recall	MAGE-CMV-Training size	CMV-test size
i- All models	0.4657	0.5355	0.4353	0.2337	4,223 H, 16,556 AI	8,097 H, 8,097 AI
ii-OpenAI-Llama	0.8503	0.9163	0.8495	0.9244	4,223 H, 4,429 AI	5,398 H, 5,397 AI

Table 15: Results of two settings for cross-CMV-MAGE setup: (i) classifier is trained on training split from MAGE-CMV data with AI-texts from all 27 models and tested on ChangeMyView dataset with all AI-texts (GPT, LLaMA, & Mistral). (ii) classifier trained on training split from MAGE-CMV with AI-texts from OpenAI & LLaMA models, then tested on ChangeMyView but with AI-texts from only GPT & LLaMA.

(from 80.84% to 53.12%), consistent with our earlier findings. However, an intriguing pattern emerges: removing morphological, psycholinguistic, and information features actually *improves* performance by up to 3.5%. This suggests these features capture domain-specific patterns that hurt generalization when both text and model domains are unseen. So, we conducted a more analysis and evaluation across these four new text domains, which we discuss next.

Separate Unseen Domains Testing each unseen domain individually (Figure 9) reveals dramatic variation in feature dependencies.

CNN (News) is extremely lexical-richness dependent. In other words, removing it drops F1 from 96.82% to 48.47%, a massive 48% collapse. Nearly all other features improve performance when removed, suggesting CNN's formal news writing style makes lexical patterns the only reliable signal.

IMDb (Reviews) has, similar to CNN, massive lexical richness dependency (-41.88%), but also benefits from removing psycholinguistic features (+5.93%) and morphological features (+2.38%). Opinion writing seems to have distinctive emotional/morphological patterns that don't generalize.

PubMed (Scientific Q&A) shows, while not as much as CNN or IMDb, strong lexical richness dependency (-23.56%) but benefits from removing surface features (+3.87%) and information features (+2.79%). The scientific Q&As domain appears to have unique surface-level patterns that mislead the classifier.

DialogSum (Conversation) is the most challenging text domain (baseline using all features: 50.24%). It shows unique behavior. Removing morphological and information features *improves* F1 by 13.47%, and 5.10% respectively. Conversely, removing semantic features hurts performance (-2.86%). Conversational text appears fundamentally different: its informal, fragmented nature means traditional linguistic features actively mislead the classifier.

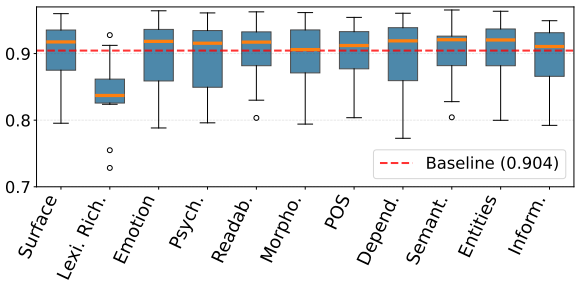
Connection to TB6 and Domain Categories These patterns align with TB6 results and reveal a clear hierarchy by domain category:

News domains (XSum, CNN) show the strongest lexical richness dependency and benefit from removing most other features. Their formal, structured writing makes vocabulary patterns paramount.

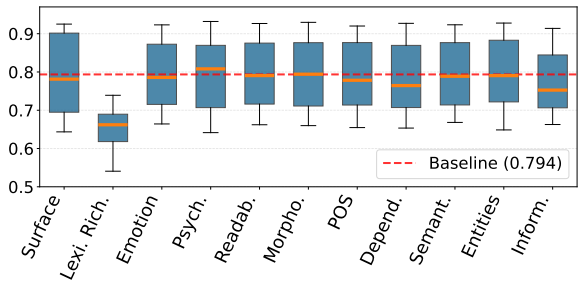
Review domains (Yelp, IMDb) also depend heavily on lexical richness but show mixed results with emotion and morphological features. Human reviews have distinctive emotional patterns that sometimes help in-domain (TB 3) but hurt generalization (Testbeds 6 and 7).

Q&A domains (ELI5, SQuAD, PubMed) vary widely. Explanatory Q&A (ELI5, PubMed) shows moderate lexical dependency and benefits from psycholinguistic features, while factual Q&A (SQuAD) shows minimal lexical dependency and struggles with most feature types.

Story domains (WP, ROCT) show moderate lexical dependency but are highly sensitive to surface and morphological features when unseen, suggesting creative writing has unique structural pat-



(a) Visualization of results distribution of ablation study on **TB3** (domain-fixed—arbitrary-model-domains) across the 11 feature areas.



(b) Visualization of results distribution of ablation study on **TB6** (unseen domains) across the 11 feature areas.

Figure 8: Distribution of F1-Macro scores across feature area ablations for in-distribution (TB3) and out-of-distribution (TB6) testbeds focusing on text domains.

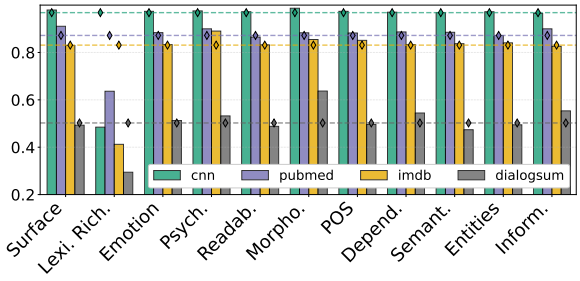


Figure 9: Visualization of results of ablation study on **TB7** using separate unseen domains (cnn, pubmed, imdb, & dialogsum) and one unseen model (gpt4). The horizontal dashed lines indicate the original baseline results for each domain, and the bars indicate the results of dropping the corresponding feature area.

terns.

Conversational domain (DialogSum) is the outlier; morphological and information features actively hurt performance, while semantic features become critical. This suggests conversational AI detection requires fundamentally different feature sets.

This domain-level analysis motivates our selection strategy for TB8 (see next section C.3): we need pairs that capture this variation, from highly lexical-dependent formal domains (XSum) to feature-sensitive conversational domains (represented by SQuAD's unusual behavior) to stable domains (TLDR). Combined with model domain variation, this ensures our 16 selected pairs comprehensively represent the feature dependency landscape.

C.3 TB8 Ablation: Selection of Domain-Model Pairs

Running ablation on all TB8 combinations (10 text domains × 7 model domains × 11 features = 770 classifiers) is computationally prohibitive. We selected 16 representative pairs based on sensitivity analysis.

Selection Methodology We measured how much feature ablation effects vary (standard deviation) in ID versus OOD scenarios (Figure 12). Text and model domains with larger changes in standard deviation show more unpredictable behavior when unseen (Figure 10).

Selected 4 text domains: XSum (highest sensitivity when unseen), WritingPrompts (high sensitivity increase), TLDR (stable behavior), and SQuAD (lowest sensitivity).

Selected 4 model domains: OpenAI (hardest to detect), LLaMA (highest sensitivity increase), BigScience (stable behavior), and EleutherAI (easiest to detect)

This yields 16 pairs covering the full spectrum from stable (SQuAD-EleutherAI) to volatile (XSum-LLaMA).

Results Figure 11 shows lexical richness remains the dominant feature across all 16 pairs, with median drops around -0.12. However, Figure 13 (heatmap showing effects exceeding ±2%) and Figure 14 (feature area patterns) reveal distinct clustering by text domain category, consistent with TB6 and TB7 patterns.

News (XSum) pairs show extreme lexical richness dependency across all model domains (-17.3% to -32.4%), with LLaMA showing the strongest dependency (-32.4%). This mirrors CNN's behavior in TB7 where removing lexical richness caused 48% performance collapse. Surface (+2.3% to +4.4%), morphological (+1.2% to +3.7%), and dependency (+3.1% to +4.0%) features consistently

Domain	Baseline	Surface	Lexi.Rich.	Emotion	Psych.	Readab.	Morpho.	POS	Depend.	Semant.	Entities	Inform.
Testbed 2												
OpenAI	.555	.551 (-.005)	.558 (+.003)	.554 (-.002)	.549 (-.007)	.553 (-.002)	.554 (-.001)	.556 (+.001)	.549 (-.006)	.556 (+.000)	.556 (+.001)	.552 (-.003)
LLaMA	.753	.752 (-.001)	.775 (+.022)	.758 (+.005)	.755 (+.002)	.755 (+.002)	.750 (-.003)	.758 (+.005)	.751 (-.002)	.757 (+.003)	.756 (+.003)	.755 (+.002)
GLM	.618	.612 (-.006)	.643 (+.025)	.617 (-.001)	.614 (-.004)	.613 (-.005)	.615 (-.003)	.615 (-.003)	.615 (-.003)	.617 (-.001)	.618 (-.000)	.616 (-.002)
FLAN-T5	.766	.752 (-.014)	.671 (-.095)	.765 (-.001)	.761 (-.004)	.765 (-.001)	.757 (-.009)	.765 (-.001)	.762 (-.004)	.764 (-.001)	.764 (-.001)	.763 (-.003)
OPT	.913	.906 (-.007)	.881 (-.032)	.913 (-.000)	.910 (-.003)	.913 (-.000)	.910 (-.003)	.912 (-.002)	.909 (-.004)	.913 (-.000)	.914 (+.000)	.909 (-.004)
BigScience	.656	.646 (-.010)	.599 (-.057)	.654 (-.002)	.647 (-.009)	.657 (+.000)	.634 (-.022)	.651 (-.005)	.650 (-.006)	.654 (-.002)	.655 (-.001)	.649 (-.007)
Eleuther	.881	.876 (-.005)	.857 (-.024)	.882 (+.001)	.878 (-.003)	.881 (+.000)	.870 (-.011)	.877 (-.004)	.876 (-.005)	.880 (-.001)	.881 (-.000)	.879 (-.002)
Testbed 5												
OpenAI	.616	.609 (-.008)	.483 (-.133)	.613 (-.003)	.607 (-.009)	.616 (-.000)	.612 (-.005)	.613 (-.003)	.609 (-.007)	.614 (-.002)	.616 (+.000)	.595 (-.021)
LLaMA	.644	.642 (-.001)	.469 (-.174)	.641 (-.002)	.637 (-.006)	.644 (+.000)	.638 (-.006)	.637 (-.006)	.634 (-.009)	.642 (-.002)	.643 (-.000)	.620 (-.024)
GLM	.496	.491 (-.005)	.458 (-.038)	.493 (-.002)	.491 (-.004)	.496 (+.000)	.493 (-.003)	.492 (-.004)	.491 (-.004)	.494 (-.002)	.496 (+.000)	.480 (-.015)
FLAN-T5	.673	.666 (-.007)	.533 (-.141)	.671 (-.002)	.669 (-.004)	.673 (+.000)	.670 (-.004)	.669 (-.004)	.667 (-.006)	.672 (-.001)	.674 (+.001)	.647 (-.026)
OPT	.714	.710 (-.004)	.583 (-.131)	.712 (-.002)	.710 (-.004)	.715 (+.000)	.711 (-.003)	.710 (-.004)	.711 (-.003)	.714 (-.001)	.715 (+.001)	.698 (-.016)
BigScience	.595	.590 (-.005)	.536 (-.059)	.593 (-.002)	.589 (-.006)	.595 (+.000)	.594 (-.001)	.592 (-.003)	.591 (-.004)	.593 (-.002)	.596 (+.001)	.575 (-.020)
Eleuther	.528	.523 (-.005)	.498 (-.029)	.525 (-.003)	.523 (-.005)	.528 (-.000)	.525 (-.003)	.524 (-.004)	.522 (-.006)	.526 (-.002)	.528 (+.000)	.510 (-.018)
Testbed 3												
CMV	.924	.918 (-.006)	.843 (-.081)	.920 (-.003)	.916 (-.008)	.923 (-.001)	.891 (-.032)	.916 (-.008)	.922 (-.002)	.922 (-.002)	.922 (-.002)	.912 (-.012)
Yelp	.804	.800 (-.005)	.728 (-.076)	.788 (-.016)	.796 (-.008)	.803 (-.001)	.794 (-.010)	.804 (-.001)	.773 (-.031)	.804 (+.000)	.800 (-.004)	.792 (-.012)
XSum	.941	.938 (-.003)	.836 (-.106)	.938 (-.003)	.940 (-.001)	.941 (-.001)	.941 (-.000)	.934 (-.007)	.943 (+.002)	.922 (-.019)	.941 (-.001)	.931 (-.010)
TLDR	.908	.907 (-.001)	.824 (-.084)	.855 (-.053)	.915 (+.007)	.909 (+.001)	.899 (-.009)	.906 (-.002)	.865 (-.044)	.909 (+.000)	.912 (+.004)	.909 (+.000)
ELI5	.873	.864 (-.009)	.755 (-.118)	.870 (-.003)	.836 (-.037)	.873 (-.000)	.864 (-.009)	.868 (-.005)	.858 (-.015)	.873 (+.000)	.872 (-.001)	.852 (-.021)
WP	.964	.960 (-.005)	.912 (-.052)	.964 (-.000)	.961 (-.003)	.963 (-.002)	.962 (-.003)	.954 (-.010)	.961 (-.004)	.965 (+.001)	.964 (-.001)	.949 (-.015)
ROCT	.928	.927 (-.000)	.868 (-.060)	.931 (+.003)	.891 (-.037)	.927 (-.001)	.913 (-.015)	.930 (+.002)	.927 (-.001)	.927 (-.000)	.925 (-.002)	.931 (+.003)
HSwag	.953	.955 (+.002)	.928 (-.025)	.957 (+.004)	.947 (-.006)	.934 (-.018)	.951 (-.002)	.953 (+.000)	.952 (-.001)	.954 (+.001)	.953 (+.000)	.947 (-.006)
SQuAD	.829	.795 (-.034)	.831 (+.002)	.810 (-.019)	.819 (-.010)	.830 (+.001)	.829 (-.001)	.816 (-.014)	.827 (-.002)	.828 (-.001)	.827 (-.002)	.827 (-.002)
SciGen	.920	.917 (-.003)	.838 (-.081)	.916 (-.003)	.918 (-.001)	.911 (-.008)	.919 (-.000)	.908 (-.011)	.916 (-.003)	.919 (-.000)	.919 (-.000)	.908 (-.011)
Testbed 6												
CMV	.866	.868 (+.002)	.739 (-.127)	.863 (-.004)	.877 (+.011)	.870 (+.003)	.845 (-.021)	.856 (-.010)	.868 (+.001)	.859 (-.007)	.868 (+.002)	.823 (-.044)
Yelp	.777	.775 (-.001)	.660 (-.116)	.766 (-.010)	.794 (+.017)	.776 (-.000)	.778 (+.002)	.764 (-.013)	.764 (-.013)	.772 (-.004)	.775 (-.001)	.740 (-.036)
XSum	.897	.913 (+.015)	.623 (-.274)	.892 (-.005)	.909 (+.011)	.900 (+.002)	.915 (+.018)	.887 (-.010)	.912 (+.015)	.897 (-.000)	.889 (-.008)	.852 (-.046)
TLDR	.704	.678 (-.026)	.617 (-.087)	.711 (+.007)	.688 (-.016)	.707 (+.004)	.703 (-.000)	.704 (+.001)	.691 (-.013)	.704 (+.000)	.713 (+.010)	.712 (+.008)
ELI5	.807	.787 (-.020)	.694 (-.113)	.805 (-.002)	.823 (+.015)	.805 (-.003)	.810 (+.003)	.793 (-.014)	.756 (-.051)	.805 (-.002)	.805 (-.002)	.765 (-.042)
WP	.883	.919 (+.035)	.676 (-.207)	.876 (-.008)	.846 (-.037)	.877 (-.006)	.887 (+.004)	.884 (+.000)	.870 (-.013)	.883 (-.001)	.888 (+.005)	.866 (-.017)
ROCT	.668	.680 (+.012)	.541 (-.128)	.683 (+.015)	.696 (+.028)	.662 (-.006)	.687 (+.019)	.682 (+.014)	.680 (+.012)	.668 (-.000)	.649 (-.020)	.704 (+.036)
HSwag	.739	.738 (-.001)	.574 (-.165)	.730 (-.009)	.738 (-.001)	.742 (+.003)	.735 (-.004)	.742 (+.003)	.765 (+.026)	.744 (+.005)	.747 (+.008)	.688 (-.051)
SQuAD	.667	.643 (-.023)	.664 (-.003)	.664 (-.002)	.641 (-.025)	.663 (-.004)	.660 (-.007)	.655 (-.012)	.653 (-.013)	.672 (+.005)	.665 (-.001)	.663 (-.004)
SciGen	.927	.925 (-.002)	.725 (-.203)	.923 (-.004)	.932 (+.005)	.927 (-.001)	.930 (+.002)	.920 (-.007)	.927 (-.000)	.923 (-.004)	.928 (+.000)	.914 (-.013)

Table 16: Complete results of the ablation study (A) where we drop a feature group at each training cycle for four different testbeds (**TB2**, **TB3**, **TB5**, & **TB6**) and the difference to baseline where the classifier is trained on the full set of features (284). The values in red indicate a decrease in performance, while green values indicate an improvement compared to the baseline result.

improve performance when removed across all models, indicating formal news writing relies almost exclusively on vocabulary patterns.

Creative writing (WP) pairs display strong lexical dependency (-11.3% to -23.0%), with LLaMA again showing the strongest effect (-23.0%). Surface features provide the largest benefit when removed (+5.8% to +7.1%), consistent across all models. Psycholinguistic features consistently hurt performance (-4.7% to -5.7%), mirroring IMDb patterns in TB7 where emotional features mislead in OOD settings. Dependency features show similar effect.

News summary (TLDR) pairs show moderate but variable lexical dependency, with strong model domain effects: minimal for TLDR-Eleuther (-0.4%) but substantial for TLDR-LLaMA (-13.2%). Unlike other text domains, TLDR uniquely benefits from entities (+0.6% to +1.4%) and information features (+0.1% to +1.0%), with BigScience and Eleuther showing the strongest gains.

Q&A (SQuAD) pairs exhibit the most distinctive and model-dependent pattern. Lexical richness shows reversed effects: beneficial to remove for SQuAD-BigScience/Eleuther (+4.6% to +5.2%) but harmful for SQuAD-LLaMA (-4.8%). This dramatic model domain difference suggests BigScience and Eleuther generate factual Q&A with less distinctive vocabulary, while LLaMA maintains lexical patterns useful for detection. All SQuAD pairs require balanced contributions from surface (-2.0% to -2.9%), psycholinguistic (-2.3% to -2.7%), and dependency (-1.2% to -1.8%) features. The consistency of these requirements across models, despite varying lexical dependencies, confirms SQuAD's behavior observed in TB6 and reinforces that factual Q&A requires fundamentally different detection strategies.

Model domain patterns emerge clearly where LLaMA consistently shows the strongest lexical richness dependency across all text domains (-4.8% to -32.4%), while OpenAI and BigScience show more text-domain-specific variation. Eleuther displays unique behavior with SQuAD (benefits from removing lexical richness) but strong dependency elsewhere, suggesting its factual generation differs

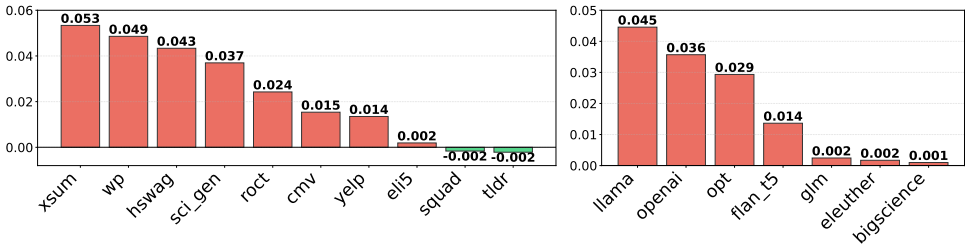


Figure 10: Change in standard deviation between unseen (**TB5** & **TB6**) and fixed (**TB2** & **TB3**) scenarios for text domains (left) and model domains (right). Red bars indicate increased sensitivity when unseen, green bars indicate decreased sensitivity. Higher absolute values represent larger shifts in model behavior across different experimental setups.

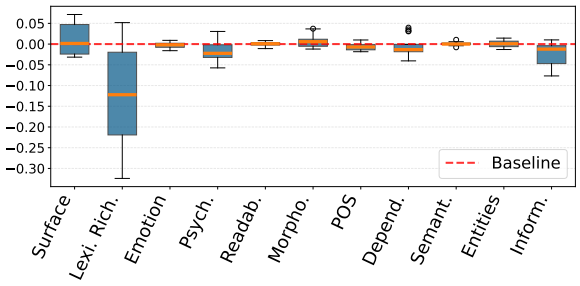


Figure 11: Leave-one-out features ablation performance variance across 16 text domain-model domain

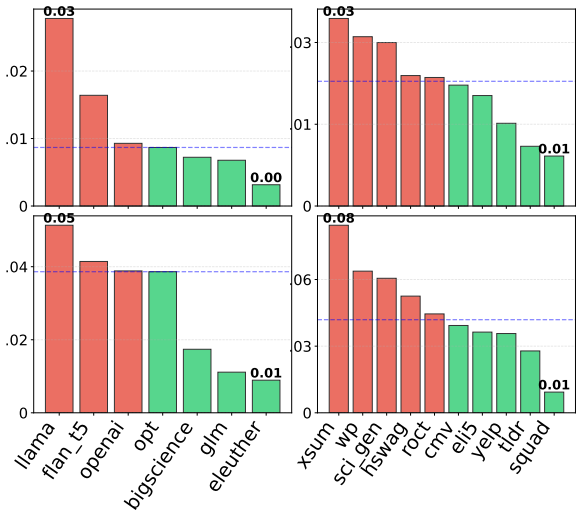


Figure 12: Results of the standard deviation between ablation study across the different testbeds and the ablation study using feature areas. Red bars indicate increased sensitivity compared to the mean, green bars indicate decreased sensitivity compared to the mean (horizontal line).

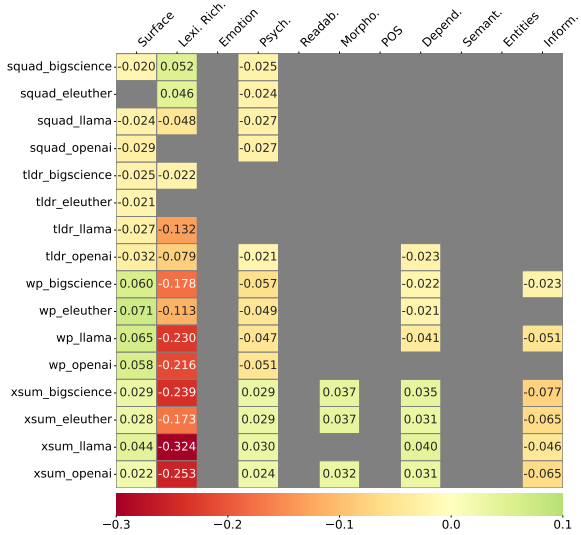


Figure 13: The difference in the performance for the 16 selected text domain-model-domain pairs from the previous study. We keep only the deltas ($\Delta = ablation_results - baseline$) of $\pm 2\%$ effect.

from its creative or news generation.

These patterns demonstrate that while text domain predicts feature dependency reliably, model domain introduces critical variation within each domain. The SQuAD pairs exemplify this: text domain determines that multiple features are required (unlike news's lexical-only strategy), but model domain determines whether lexical richness helps or hurts. This nuanced interaction reinforces our selection methodology: pairs showing high standard deviation changes between ID and OOD capture these complex dependencies essential for understanding real-world detection challenges.

C.4 Lexical Richness Results in Unseen Domain-Model Pairs (TB8)

Our ablation studies revealed lexical richness as the most critical feature group across all testbeds.

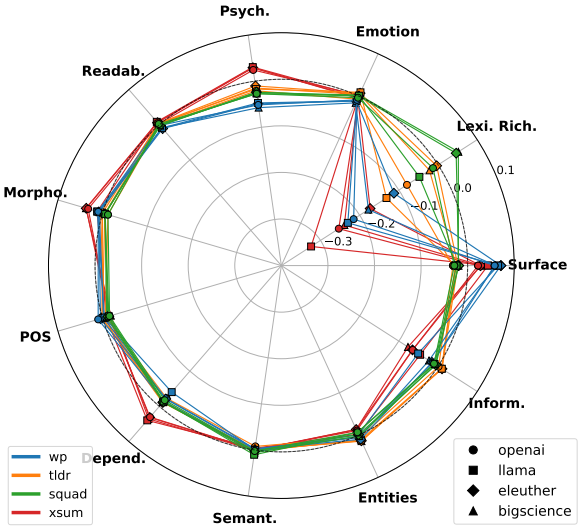


Figure 14: The difference in the performance for the 16 selected text-domain-model-domain pairs from the previous study using the 16 text-domain-model-domain pairs. The text domains separated by colors and and the model domains by markers/ shapes.

The cumulative ablation on TB7 (unseen domains and model) showed a surprising result: using *only* lexical richness features outperformed the full 284-feature baseline by +14.29% F1. This motivated us to investigate whether this pattern holds for specific unseen domain-model pairs in TB8.

Methodology Since we cannot run the cumulative ablation on all testbeds and all settings, we train the models using only lexical richness features. We report results for TB8 below (with comparison to results of TB1.1). For the 70 text domain-model domain pairs in both testbeds (TB1.1 & TB8), we trained classifiers using only the lexical richness feature area and compared performance to the baseline (trained on all 284 features). The delta values (Figure 15) show the F1-Macro difference between lexical-only and baseline classifiers.

Results of OOD Performance (TB8) Positive values (green in heatmap) indicate pairs where lexical richness alone outperforms the full feature set: meaning the other 281 features actually introduce noise. Negative values (red) indicate traditional behavior where more features help. We distinguish two patterns in Figure 15b to report the text domain patterns.

C.4.1 Text domain patterns

ROCT shows the strongest improvement when using only lexical features, with gains across all model domains (+10.1% to +15.3%). Story generation appears to have such distinctive lexical patterns that other linguistic features add confusion.

WP also benefits substantially from lexical-only classification for most models (+4.2% to +16.3%), except with OPT, LLaMA, and GLM which show opposite behavior with drops of -20.0%, -16.6%, and -2.0% respectively. This suggests some models produce creative text with additional non-lexical signals.

Yelp and CMV (opinion/review domains) generally improve with lexical-only features, particularly for OpenAI (+6.97% CMV, +3.75% Yelp) and FLAN-T5 models.

SciGen, SQuAD, and HSwag consistently perform worse with lexical-only features (all negative values except for the pair of HSwag-OPT), indicating these domains require the full feature set. Scientific writing and factual Q&A have more subtle linguistic patterns beyond vocabulary choice.

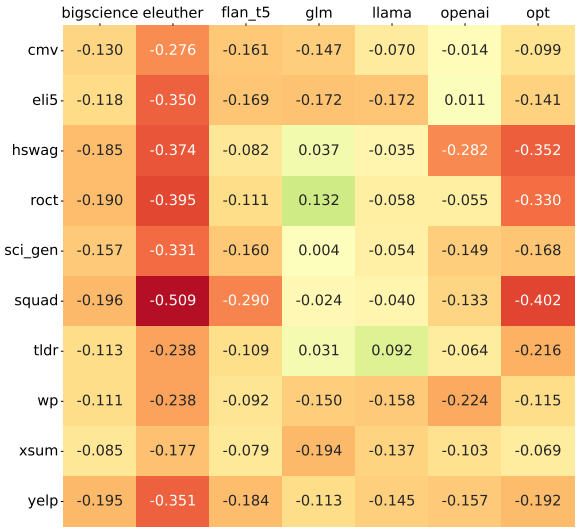
C.4.2 Model domain patterns

OpenAI domain shows the most consistent improvement with lexical-only features (7/10 text domains positive), with an average delta of +0.0528 across text domains.

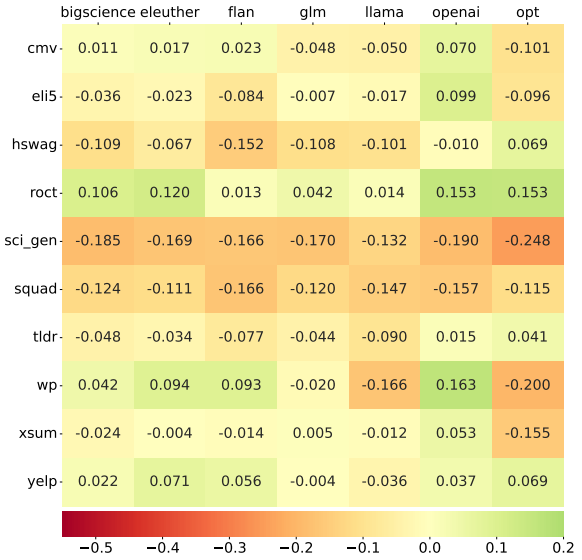
BigScience, Eleuther, Flan, and OPT domains show mixed results depending on the text domain. However, the OPT model domain shows the largest shift and text-domain dependence in the results. In text domains like SciGen, WP, and XSum, OPT models does not benefit from training on the lexical richness features, but the opposite happens in ROCT domain. This behavior can be also observer in the other model domains (BigScience, Eleuther, and Flan) with some difference. This suggest that these models generation of text depends on the text domain to generate lexically rich texts.

GLM and LLaMA model domains show more stable results, where both seem to perform better with full features sets in most of the text domains with few exceptions (ROCT-GLM, ROCT-LLaMA, and Xsum-GLM with performance improvements of +4.2%, +1.4%, and +0.5% respectively).

Comparison to In-Distribution Performance (TB1.1) The striking difference between in-distribution (TB1.1) and out-of-distribution (TB8) performance reveals when lexical features alone suffice versus when the full feature set is necessary. In TB1.1, where both text and model domains are seen during training, lexical-only features consistently underperform the baseline across nearly



(a) Results of **TB1.1** domain-model pairs. Positive values (green) indicate lexical richness alone outperforms all 284 features; negative values (red) indicate the full feature set is superior.



(b) Results of **TB8** domain-model pairs. Positive values (green) indicate lexical richness alone outperforms all 284 features; negative values (red) indicate the full feature set is superior.

Figure 15: Performance difference (F1-Macro) between lexical-richness-only and full-feature classifiers.

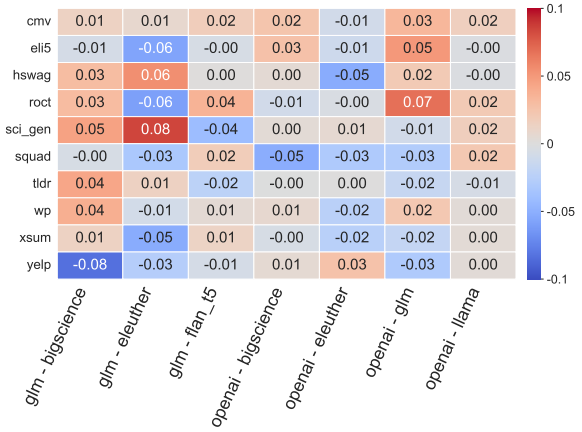


Figure 16: Heatmap of pairwise Pearson's r of TTR distributions between selected model family pairs across text domains. The left columns show GLM paired with smaller models, while the right columns show OpenAI paired with models of varying scale. Cell values indicate the Pearson's r between the TTR distributions of the two model families within each text domain.

all 70 pairs (Figure 15a). The worst drops occur with Eleuther models (average -0.327), particularly on SQuAD (-50.9%). This indicates that when training and test data match, classifiers effectively leverage text domain-specific and model-specific patterns from all feature areas. However, in TB8 where both text and model domains are unseen, the pattern reverses for many pairs. Notably, ROCT flips from negative in TB1.1 (average -0.217) to strongly positive in TB8 (average +0.106), and OpenAI models shift from negative (average -0.141 in TB1.1) to positive (average +0.053 in TB8). Additionally, GLM exhibits a unique reversal pattern: it's shows positive values in TB1.1 for multiple text domains (ROCT +0.132, HSwag +0.037, TLDR +0.031, SciGen +0.004), yet these same domains become negative in OOD setting. The pattern flips completely between TB1.1 and TB8: what works for seen GLM fails for unseen GLM, and vice versa, almost across all text domains. This suggests that while domain-specific features help when available, they become misleading in cross-generalization scenarios where lexical richness provides more robust, generalizable signals.

C.4.3 Feature Distributions Analysis

Given the substantial impact of lexical richness features across all testbeds, we analyzed the distributions of its three features: TTR (Type token ratio of the text: n_types/n_tokens), global token hapax legomena (Number of hapax legomena -tokens that occur only once- in the entire corpus in the text instance), and lexical density (Lexical density of the text: $n_lexical_tokens/n_tokens$).

Distribution Analysis of Lexical Richness Features The violin plots (Figure 18) and histograms (Figure 17) reveal distinct distributional patterns across the three lexical richness features. Human-

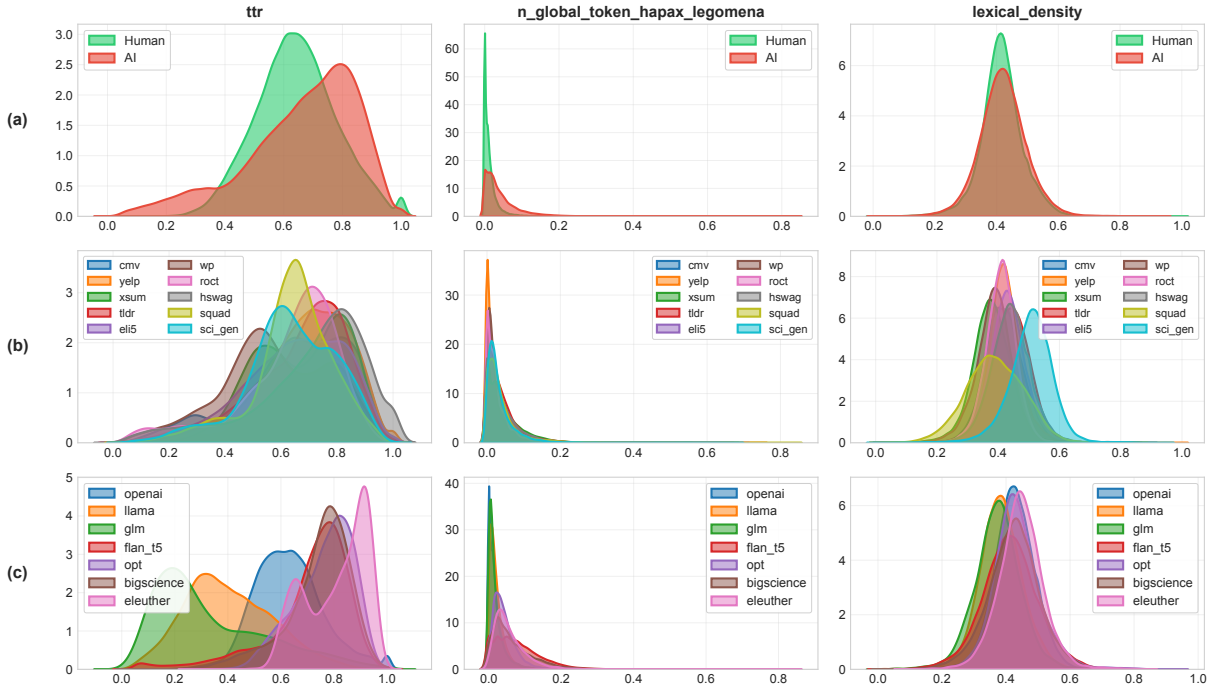


Figure 17: Density distributions of lexical richness features (TTR, hapax legomena, lexical density) across (a) Human vs AI labels, (b) text domains, and (c) AI model domains. Overlapping histograms show the discriminative power of these features across different groupings.

written text consistently exhibits higher values across all three features compared to AI-generated text (row a), indicating greater lexical diversity in human writing. The TTR distribution shows human text centered around 0.6 while AI text clusters around 0.8, demonstrating that human authors use a more varied vocabulary relative to text length. Hapax legomena-words appearing only once in a text-are substantially more frequent in human writing, with distributions showing clear separation between the two classes. This pattern holds across nearly all text (row b) and model (row c) domains, although the magnitude of separation varies, suggesting that lexical richness features are a robust discriminative signal for AI-generated text detection, which supports our findings.

Domain and Model-Specific Variations While the human-AI separation is consistent across conditions, notable variations emerge when examining specific text and model domains. Domain-level analysis (row b) shows that certain text domains like SciGen and SQuAD exhibit tighter distributions and reduced human-AI separation, particularly in hapax legomena counts. Model domain comparisons (row c) reveal that some AI-text generators, particularly OpenAI, GLM, and Llama models, produce text with hapax legomena counts values closer to human baselines compared to other model domains like Flan-T5 or OPT, which show more pronounced deviations in TTR and lexical density. These variations explain why lexical richness features, despite being among the most discriminative overall, show differential effectiveness across the experimental testbeds, with performance varying more substantially in scenarios involving specific or unseen text-model domains combinations where the distributional overlap is greater.

D Qualitative Examples

To complement our quatitative analysis, we present qualitative examples that illustrate how the extracted linguistic features manifest in human-written versus AI-generated text. We select two feature areas–Emotion and Lexical Richness– based on their performance and clear distinction in both texts, and choose the domain-model pairs that best highlight the contrast identified in our experiments.

D.1 Emotion Features — Domain: Yelp, Model: OpenAI text-davinci-003

We select the Yelp domain paired with OpenAI's `text-davinci-003` for the emotion feature visualization (Figure 19), as Yelp reviews are inherently opinion-driven and exhibit a wide range of emotional expression among the text domains.

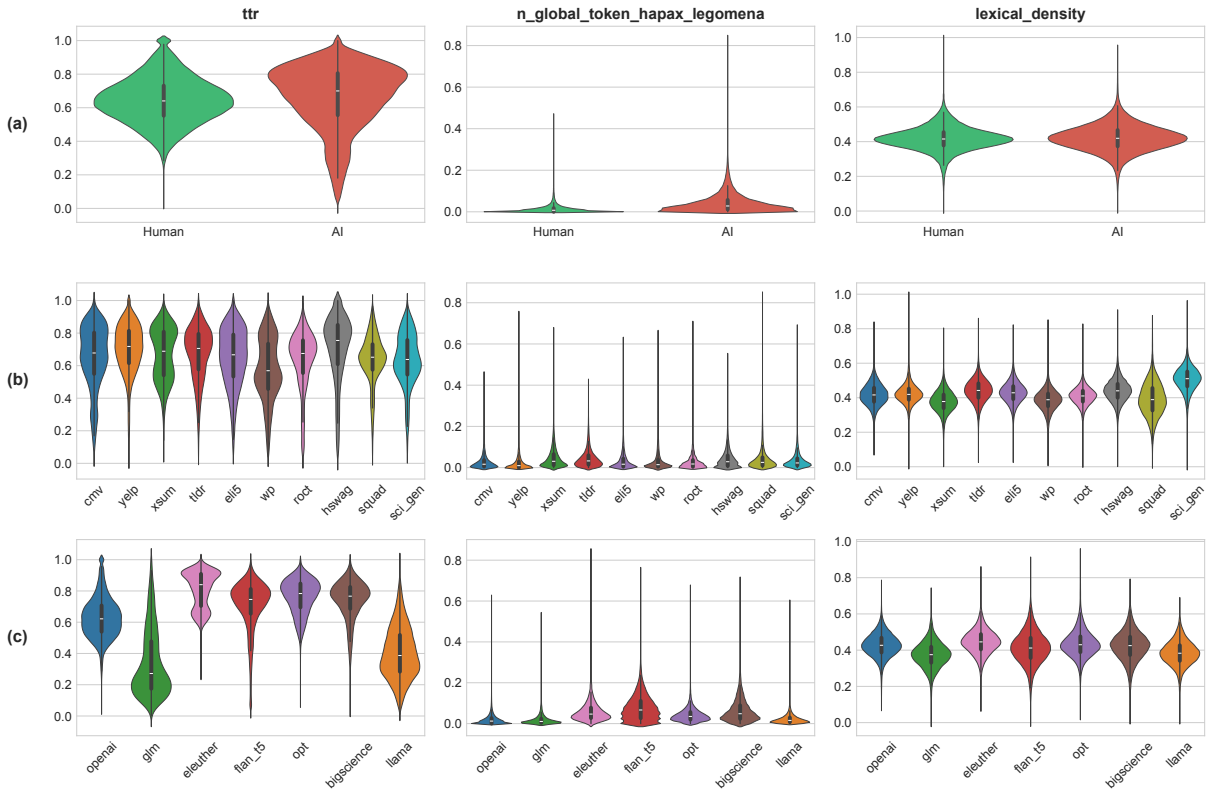


Figure 18: Violin plots for the three lexical richness features: TTR (left), hapax legomena (middle), and lexical density (right). Each row displays distributions across Human vs AI, text domains, and model domains, revealing both central tendency and distributional shape.

For each sample, we display two emotion intensity dimensions—anger (red) and disgust (blue)—where darker shading indicates higher intensity scores from the NRC Affect Intensity Lexicon(Mohammad, 2018b). Valence is encoded background highlighting: green indicates high valence (> 0.66) and pink indicates low valence (< 0.33) based on the NRC VAD Lexicon (Mohammad, 2018a). Aggregate statistics (avg_intensity, avg_valence, n_low/high_valence) are shown above each sample.

The examples reveal a consistent pattern: human-written Yelp reviews exhibit higher anger and disgust intensities with more low-valence tokens, reflecting genuine frustration and emotional variability. In contrast, AI-generated samples tend toward elevated valence with fewer or no anger/disgust signals, even when the generated content describes negative experiences that are parallel to human text.

D.2 Lexical Richness — Domain: XSum, Models: LLaMA & OpenAI

We select XSum paired with LLaMA (-7B,-13B) and OpenAI (text-davinci-003) for the lexical richness visualization (Figure 20), as these pairs (combinations of domain and models) exhibited one of the largest performance drops in TB8 when lexical richness features were removed (cf. Figure 13), and the spider plot analysis (Figure 14) confirmed that lexical richness is the dominant feature area for these pairs. Furthermore, XSum (news summarization) produces structurally diverse texts where lexical patterns differ markedly between human and AI writing.

Each token is annotated along three dimensions: *italic* marks first occurrences (unique types contributing to TTR), **bold** marks lexical tokens (nouns, verbs, adjectives, and adverbs contributing to lexical density), and underline marks hapax legomena (tokens appearing only once in the text). Aggregate statistics (ttr, lexical_density, n_hapax_legomena, n_tokens, n_types) are displayed above each sample.

The examples illustrate a recurring pattern: human-written texts tend to be longer with lower TTR due to natural repetition, yet contain more hapax legomena in absolute terms. AI-generated continuations are typically shorter but exhibit higher TTR and lexical density, suggesting a more concentrated but less varied vocabulary distribution.

E Distributional Similarity of Text and Model Domains

E.1 Wasserstein Distance of Text Domains

The clustered heatmap (Figure 21) reveals two distinct groups of text domains pairs based on their Wasserstein distances across the top 10 most discriminative features. The first group, appearing on the left side of the heatmap, includes pairs involving domains such as ROCStories, HellaSwag, WritingPrompts, and TLDR, which exhibit notably higher distributional divergence, particularly for surface-level features such as `raw_sequence_length`; `n_tokens`, `n_syllables`, and `n_lemmas`. This is consistent with the substantial OOD performance drops for ROC-Stories (-26.0%), HellaSwag (-21.4%), and TLDR (-20.5%) in TB6 (See Table 4), suggesting that their distinct linguistic distributions make cross-domain generalization harder. The second group, on the right side, includes pairs involving CMV, XSum, ELI5, SQuAD, and Yelp, which show lower distributional divergence overall, aligning with their comparatively smaller OOD performance drops.

Interestingly, `entropy` and `n_PRON_Number_Sing` exhibit a different clustering pattern from the surface features, showing higher divergence for pairs involving Yelp and SQuAD domains where emotion and psycholinguistic features were found to be more informative in the ablation studies. This suggests that while surface features largely drive the between-domain distributional differences, other feature areas capture complementary domain-specific signals that further explain the variability in cross-domain generalization observed in the main experiments.

E.2 Pearson's r Calculations of Model Domains

Building on the distributional analysis of lexical richness features in Section C.4.3 (See Figures 17 and 18), which revealed that TTR exhibits the most pronounced separation between human and AI-generated text across both model and text domains, we compute pairwise Pearson's r between model domain (family) distributions on the TTR feature across text domains. Specifically, for each pair of model families, we compute the correlation between their TTR feature distributions within each text domain, yielding a distribution of r values across the ten MAGE text domain. We group the model family pairs into three categories following the clustering observed in the ablation studies (cr. Section 4.2): *Smaller–Smaller* (e.g., FLAN-T5 vs OPT), *Smaller–LlaMA/OpenAI* (e.g., Eleuther vs. LLaMA), and *LLaMA–OpenAI*.

Figure 22 presents the Person's r distributions across the 21 model family pairs. Overall, the r values are low and centered around zero suggesting that model families do not exhibit strongly correlated TTR patterns across text domains. This is consistent with the model domain ablation findings (Section 4.2), where lexical richness features displayed opposing effects for different model families in both ID and OOD settings: improving performance for some (e.g., FLAN-T5, OPT, BigScience, Eluether) while degrading it for others (e.g., OpenAI, LLaMA, GLM). The near-zero correlations observed here suggest that these model families occupy distinct regions of the TTR feature space, which explains why a single classifier trained on the combined feature space struggles to generalize across model domain in OOD settings.

Notably, the *Smaller—Smaller* group (blue) displays slightly more variance in r values compared to the other two groups, with some pairs such as *glm–eleuther* showing larger mid-spread ranges. This variability within the smaller model group aligns with observation that smaller models display more salient and model-specific linguistic signatures (Section 4.2), making their TTR distributions less consistently correlated across text domains. In contrast, the *LLaMA-OpenAI* comparison (green) shows r values tightly concentrated around zero with minimal variance, corroborating the finding the these two large model families share similar lexical richness patterns, which explains the performance improvement observed for OpenAI in OOD settings where LLaMA data is present in training.

Figure 16 presents Pearson's r values for selected model family pairs that reflect the contrasts identified in the ablation analysis (Section 4.2): GLM paired with smaller models (left), and OpenAI paired with models of different scale (right). Overall, the near-zero correlations confirm that model families occupy distinct regions of the TTR feature space across text domains. Among the GLM pairs, `glm-eleuther` shows the largest deviations, with positive correlations for SciGen ($r = 0.08$) and negative correlations for ELI5 and ROCT ($r = -0.06$ each). Similarly, `glm-bigscience` exhibit more negative correlations for Yelp ($r = -0.08$). These domain-specific deviations are con-

sistent with the ablation findings, where GLM exhibited model-specific linguistic signature that vary substantially across text domains, particularly for story generation (ROCT) and opinion domains (Yelp, ELI5).

Among the OpenAI pairs, `openai-glm` shows the most notable deviation for ROCT ($r = 0.07$) and ELI5 ($r = 0.05$), while all other pairs remain relatively close to zero. The fact that ROCT consistently shows the largest deviations across both GLM and OpenAI pairs aligns with the OOD results (Table 4), where ROCT exhibited the largest performance drop (-26.0%) in unseen domain settings, suggesting that story generation elicits the most domain-specific TTR behavior across model families. Finally, `openai-llama` shows uniformly near-zero correlations ($|r| \leq 0.02$), indicating that despite both being large model families, their TTR distributions do not co-vary — consistent with their contrasting OOD behaviors where OpenAI improves (+6.1%; Table 3) while LLaMA degrades (-11.0%).

F Use Of AI Assistants

In this work, GitHub Copilot[14] (version as of November 2025) was used as a code completion/suggestion tool. Additionally, AI-assisted writing tools like Grammarly[15] (accessed November 2025 - January 2026) have been used for spelling checks and grammar corrections.

[14] https://github.com/features/copilot
[15] https://app.grammarly.com/

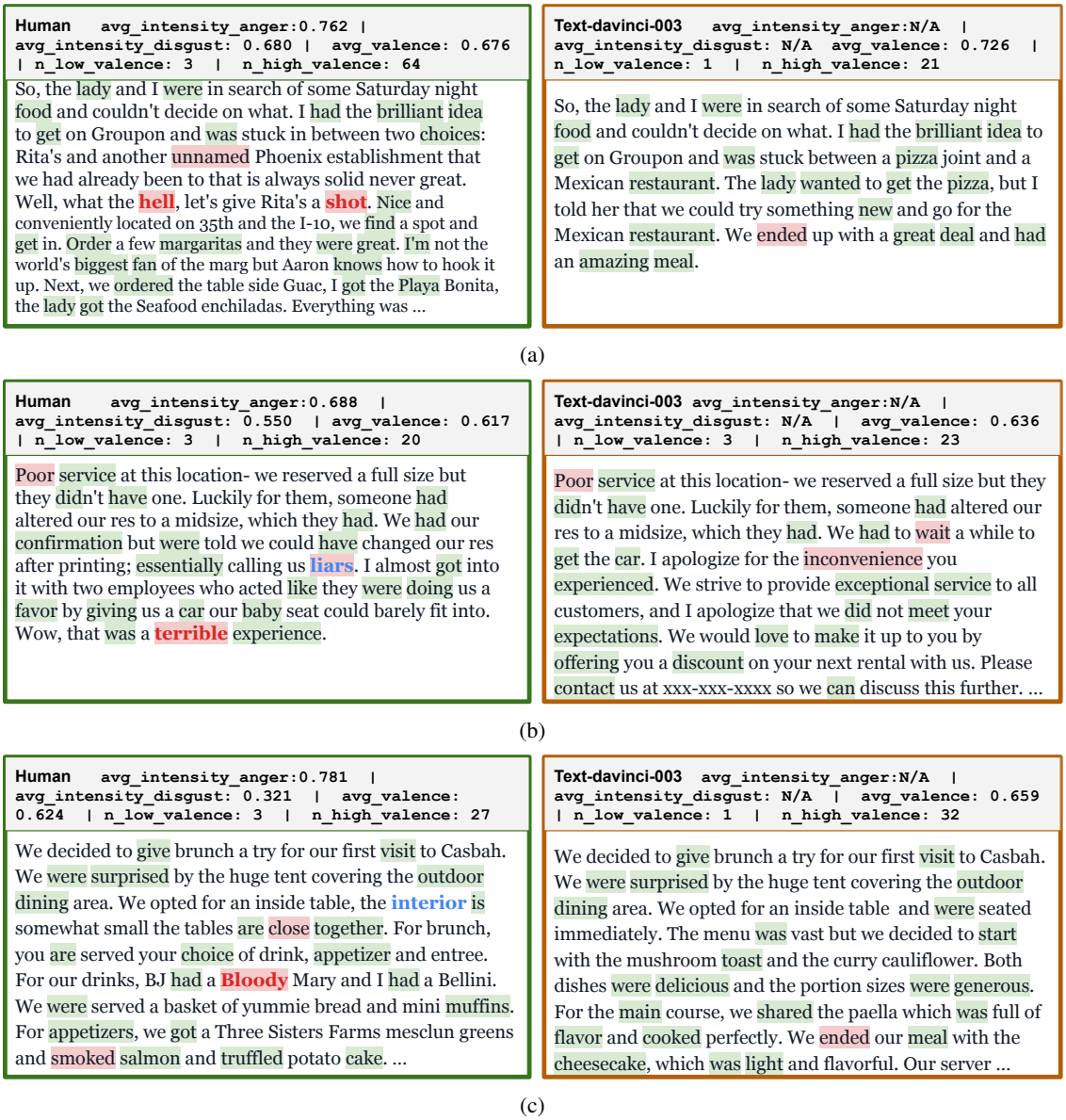
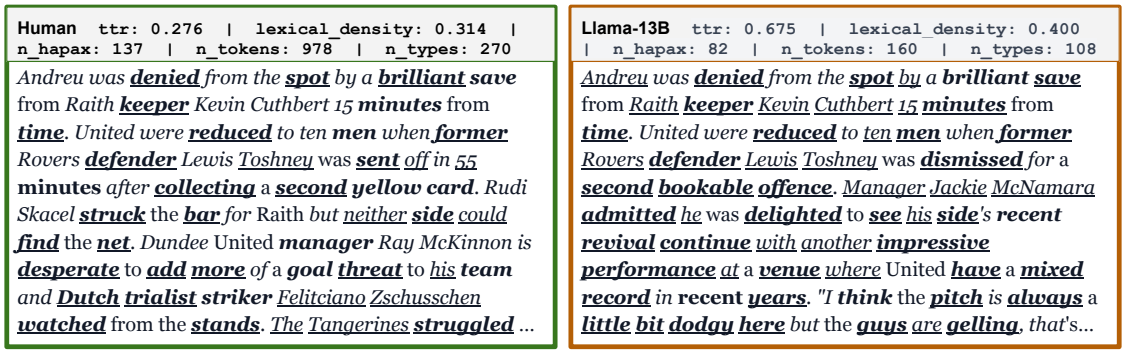


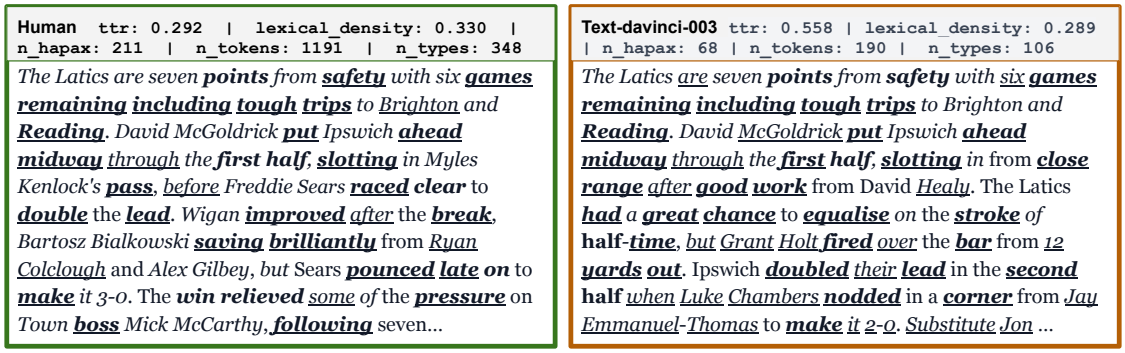
Figure 19: Parallel human–AI text pairs from the Yelp domain (AI model: Text-davinci-003) illustrating differences in a few selected features from emotion area. Visual markers: **red bold text** = anger intensity, **blue bold text** = disgust intensity, green highlight = high valence (> 0.66) and pink highlight = low valence (< 0.33). feature values (average anger and disgust intensity, average valence, and high/low valence word counts) are shown above each text. Emotion intensity is derived from the NRC Affect Intensity Lexicon and valence from the NRC VAD lexicon, following the `elfen` toolkit's extraction procedure. Some texts are truncated for space; ellipses (...) indicate omitted content.



(a) Human (Left) vs. LLaMA-7B (right).



(b) Human (left) vs. LLaMA-13 (right).



(c) Human (left) vs. Text-davinci-OO3 (right).

Figure 20: Parallel human-AI text pairs from the XSum domain illustrating differences in lexical richness features. Each pair shares the same opening words, with the AI text generated as a continuation. Visual markers: *italic*= first occurrence of a token (unique type, contributing to TTR), **bold** = lexical token (noun, verb, adjective, or adverb, contributing to lexical density, and underline = hapax legomenon (taken appearing only once in the text). Feature values (TTR, lexical density, hapax count, token count, and type count) are shown above each text. Texts are truncated for space; ellipses (. . .) indicate omitted content.

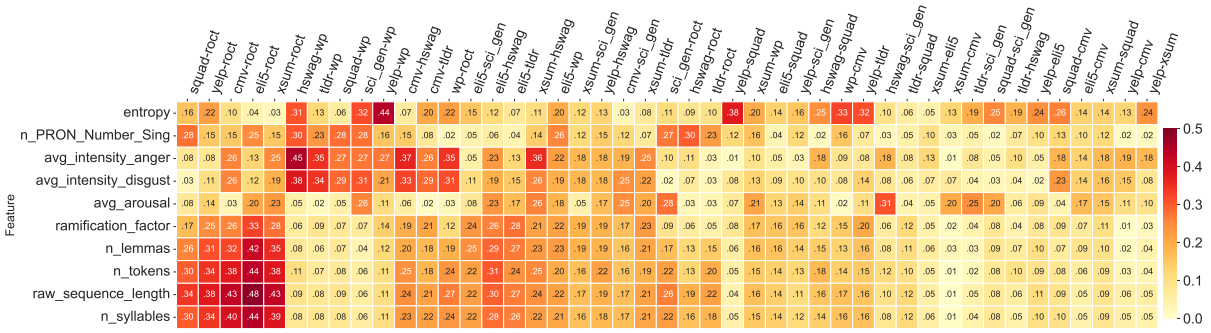


Figure 21: Clustered heatmap of pairwise Wasserstein distances for the top 10 most discriminative linguistic features across all human text domain pairs. Rows represent features that most consistently appear in the top 10 most discriminative features across domain pairs, ranked by their frequency of occurrence. Columns represent each pair of domains. Higher values (darker cells) indicate greater distributional divergence between the two domains for that feature, suggesting that the feature captures domain-specific linguistic patterns. Lower values (lighter cells) indicate similar distributions across domains, suggesting domain-invariant characteristics

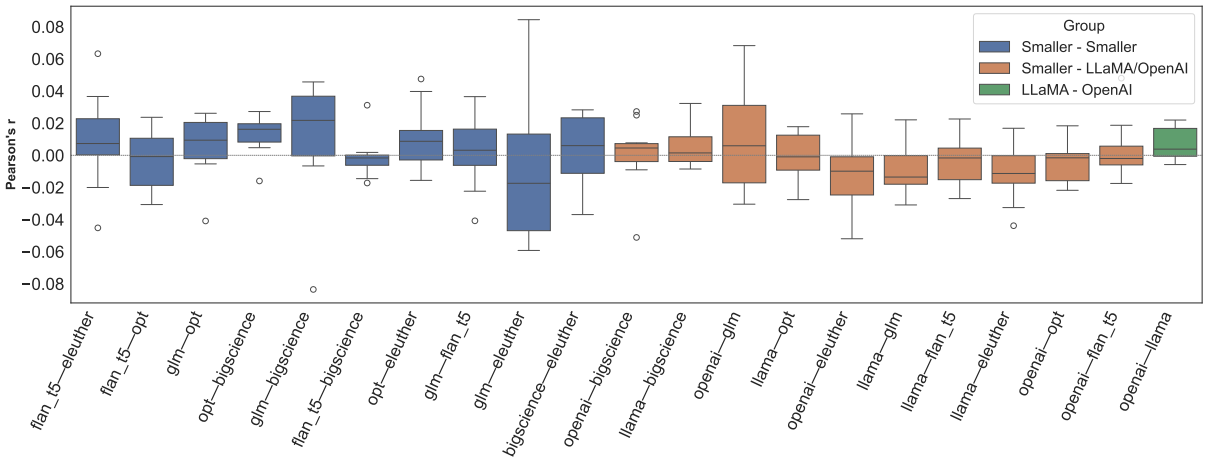


Figure 22: Pairwise Pearson's r of TTR feature distributions across model family pairs, with distributions computed over the ten text domains. Pairs are grouped into three categories: *Smaller–Smaller* (blue; e.g., FLAN-T5 vs. OPT), *Smaller–LLaMA/OpenAI* (orange; e.g., Eleuther vs. LLaMA), and *LLaMA–OpenAI* (green). TTR was selected as the focal feature given its pronounced human–AI distributional separation observed in Section C.4.3.